

Hypothesis Testing

**Course: Foundations of Statistics for Data Analytics and
Machine Learning Using Excel**

@ DV Data Analytics, Bangalore

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Review: Standard Normal Distribution and CLT

Standard Normal Distribution

The standard normal distribution has the following functions:

Probability Density Function (PDF):

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Cumulative Distribution Function (CDF):

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz.$$

where $z \in (-\infty, \infty)$

- This gives the area under the curve.

General Normal Distribution: $X \sim \mathcal{N}(\mu, \sigma^2)$

- The CDF $\Phi(z)$ still refers to the standard normal
- To compute probabilities, we standardize:

$$Z = \frac{X - \mu}{\sigma}$$

Common Probabilities (after standardization)

$$P(X < b) = \Phi\left(\frac{b - \mu}{\sigma}\right)$$

$$P(a < X < b) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right)$$

Standard Normal Distribution: $X \sim \mathcal{N}(0, 1)$

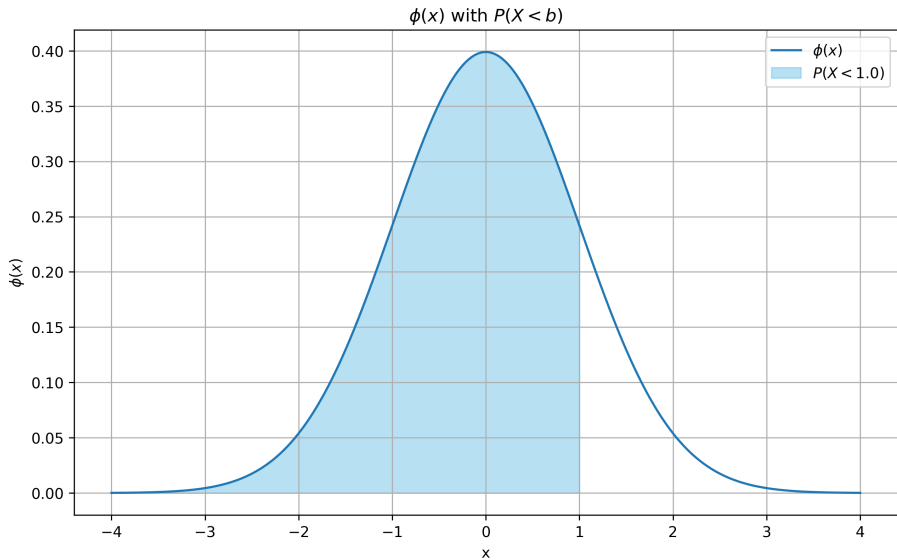
- $\mu = 0, \sigma = 1$
- The cumulative distribution function (CDF) is denoted by $\Phi(x)$

Common Probabilities

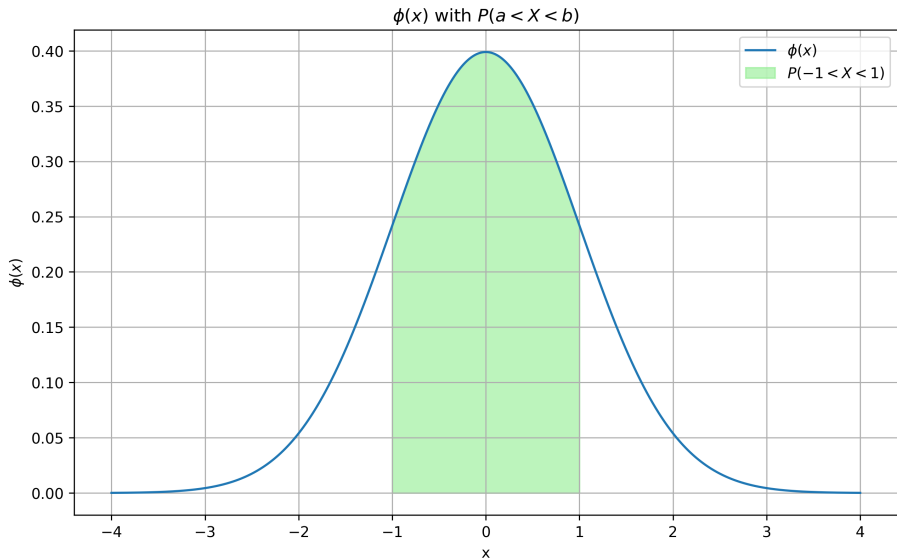
$$P(X < b) = \Phi(b)$$

$$P(a < X < b) = \Phi(b) - \Phi(a)$$

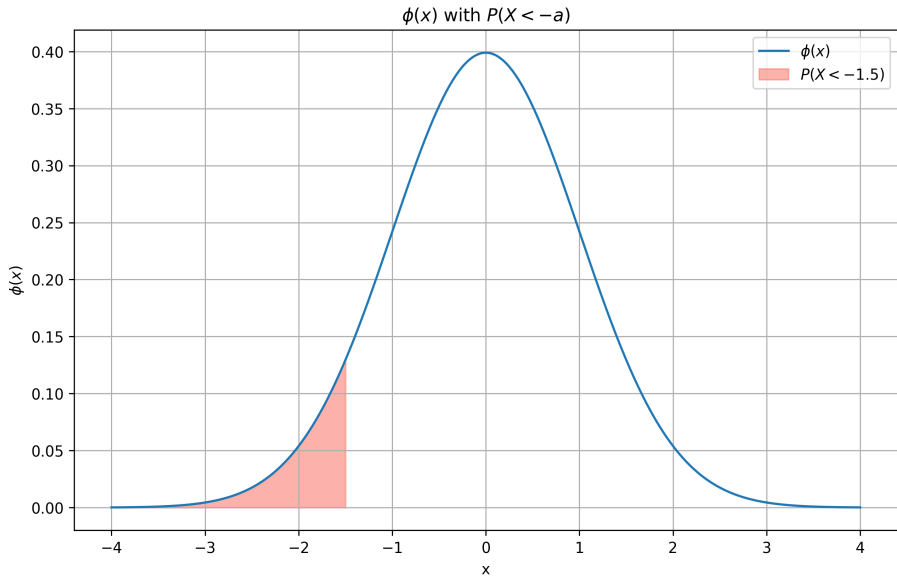
$$\phi(x), \quad x = \frac{b - \mu}{\sigma}$$



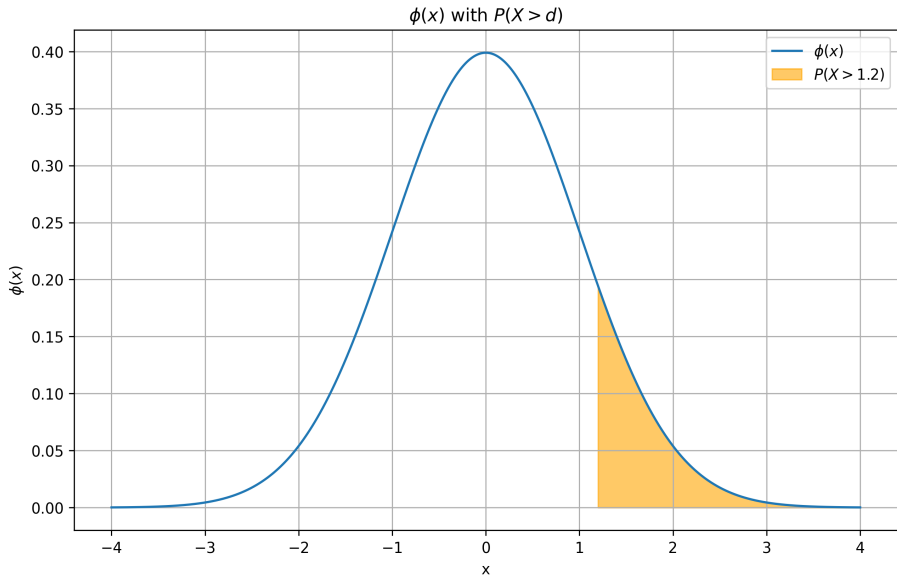
$$\phi\left(\frac{b - \mu}{\sigma}\right) - \phi\left(\frac{a - \mu}{\sigma}\right)$$



$$\phi(x)$$

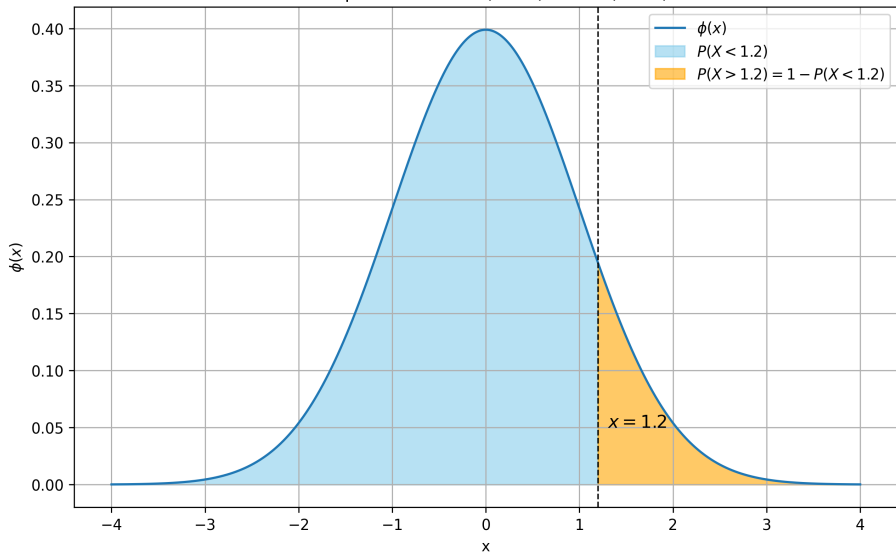


$$\phi(x)$$



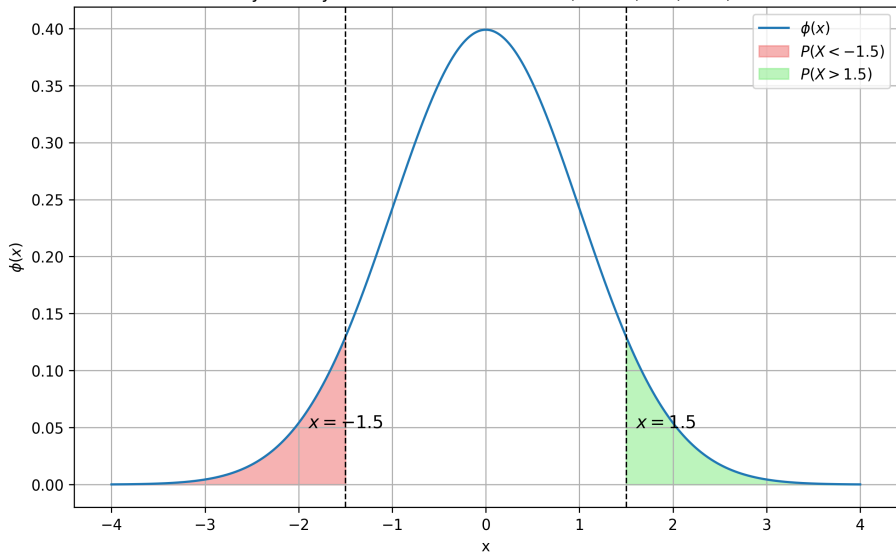
$$\phi(x)$$

Complement Rule: $P(X > d) = 1 - P(X < d)$



$$\phi(x)$$

Symmetry of the Standard Normal: $P(X < -a) = P(X > a)$



Population and Sample

What is the average number of cups of coffee per day consumed by college students in the U.S.?

- Population: All college students in the U.S.
- Parameter of interest: Average daily coffee consumption

Why We Sample

- It's impractical to survey the entire population.
- We take a random sample to estimate the population mean.

Example:

Randomly select 30 students and ask how many cups of coffee they drink per day.

Sample Data (Example) [2, 0, 1, 3, 2, 1, 2, 4, 0, 1, 2, 3, 1, 2, 2, 1, 5, 1, 0, 3, 2, 1, 2, 0, 4, 3, 2, 1, 2, 1]

- Sample size: $n = 30$
- Sample mean: (computed visually)

Each Response is a Random Variable

- Let X_i = cups of coffee consumed by the i -th student.
- Each X_i is a random variable:
 - Depends on which student is randomly chosen
 - Value varies with each sample
- The sample mean $\bar{X}$ is also a random variable.

Summary Table:

Concept	Example
Population	All U.S. college students
Random Variable X_i	Coffee cups consumed by student i
Sample	30 values like [2, 0, 1, ..., 1]
Random Process	Selecting students at random
Why It's Random	Different students give different values

Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be a random sample of **size** n from a population with:

- Mean μ
- Variance σ^2

Define the sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Then:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

As $n \rightarrow \infty$, the standardized sample mean converges in distribution to a standard normal distribution.

Mean and Variance of the Sample Mean

Expected value of $\bar{X}$:

$$\mathbb{E}[\bar{X}] = \mu$$

Variance of $\bar{X}$:

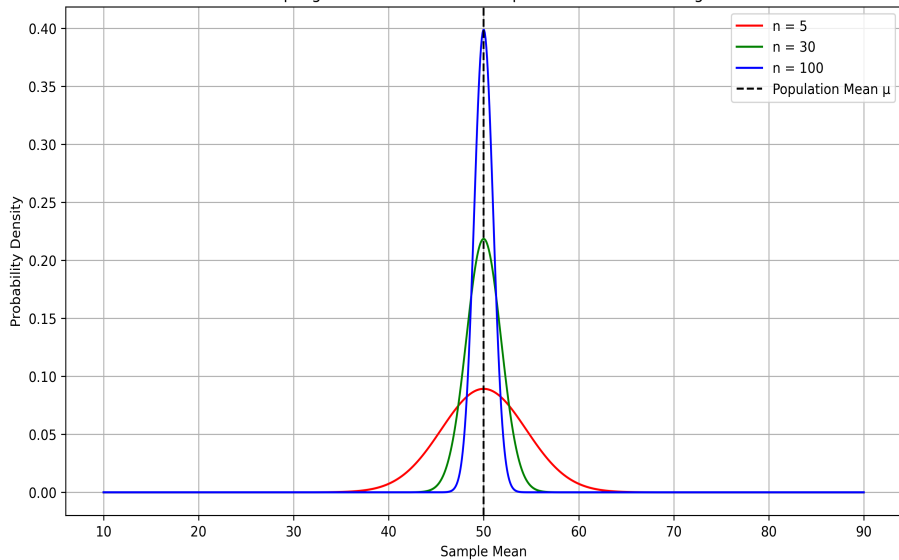
$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

Interpretation:

- The mean of the sample mean is the same as the population mean.
- The variance of the sample mean decreases as n increases.
- Larger samples $\rightarrow$ less variability in $\bar{X} \rightarrow$ tighter distribution.

Visual Intuition

Sampling Distributions of the Sample Mean with Increasing n



CLT Assumptions: Mean and Variance

The classical Central Limit Theorem assumes a sequence of i.i.d. random variables:

$$X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{a distribution with mean } \mu \text{ and variance } \sigma^2$$

i.i.d. means:

- **Independent:** Values are not related — knowing one tells you nothing about another.
- **Identically distributed:** All variables come from the same distribution.

Therefore, for CLT to apply:

- Each X_i must have the **same mean** μ
- Each X_i must have the **same variance** σ^2

Are There Exceptions? Yes!

There are **generalized versions** of the CLT that allow:

- Non-identical means or variances
- Random variables that are not identically distributed
- Some types of dependence (with conditions)

Example:

The *Lindeberg-Feller CLT* relaxes the i.i.d. assumption and allows for varying distributions, under certain regularity conditions.

These versions are more advanced but useful in real-world scenarios!

Point and Interval Estimation

What is a Point Estimate?

- A **point estimate** is a single numerical value (**Obtained From a Sample**) used to estimate a population parameter.
- Examples:
 - Sample mean ($\bar{x}$) estimates the population mean (μ).
 - Sample proportion ($\hat{p}$) estimates the population proportion (p).
 - Sample standard deviation (s) estimates the population standard deviation (σ).
- **Limitation:** Point estimates do not indicate the accuracy or reliability of the estimate.

Properties of a Good Estimator

- **Unbiasedness:** The expected value of the estimator equals the true parameter value.
- **Consistency:** As sample size increases, the estimator approaches the true parameter.
- **Efficiency:** The estimator has the smallest possible variance.
- **Sufficiency:** Uses all available information in the data.

What is an Interval Estimate?

- An interval estimate provides a range of values that is likely to contain the population parameter.
- It is expressed as:

$$\text{Point Estimate} \pm \text{Margin of Error}$$

- Example: A 95% confidence interval for the population mean.

Steps

- Take a random sample.
- Compute the statistic (**Point estimate**).
- Provide interval for this point estimate (**Interval estimate**).

Sampling from a Normal Population

Suppose $X_1, \dots, X_n$ is a sample from a normal population with unknown mean μ and known variance σ^2 .

It has been shown that the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

is the maximum likelihood estimator for μ .

We use confidence intervals to capture the likely range of μ based on the sampling distribution of $\bar{X}$.

Standardizing the Sample Mean

Since $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$, it follows that:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

So,

$$P\left(-1.96 < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < 1.96\right) = 0.95$$

Multiplying all terms by $\sigma/\sqrt{n}$:

$$P\left(-1.96 \cdot \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < 1.96 \cdot \frac{\sigma}{\sqrt{n}}\right) = 0.95$$

Rearranging the Inequality

Rewriting the inequality gives:

$$P\left(\bar{X} - 1.96 \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96 \cdot \frac{\sigma}{\sqrt{n}}\right) = 0.95$$

That is, with 95% confidence, we assert that the true mean μ lies within:

$$\bar{X} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

Final Confidence Interval Expression

Therefore, the 95% confidence interval for the population mean μ is:

$$\left(\bar{X} - 1.96 \cdot \frac{\sigma}{\sqrt{n}}, \quad \bar{X} + 1.96 \cdot \frac{\sigma}{\sqrt{n}} \right)$$

This interval is called the **95% confidence interval estimate of μ** .

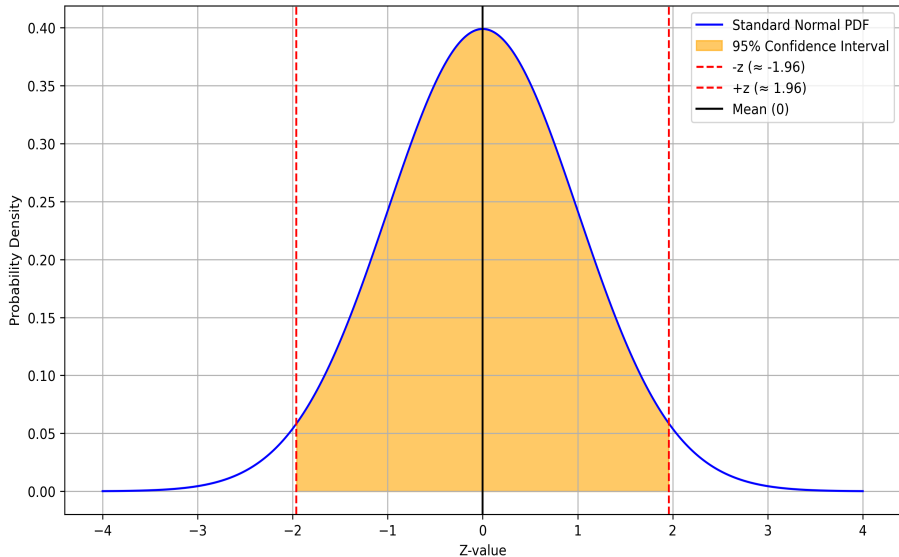
- The normal distribution curve (PDF) has a total area of 1, representing 100% probability.
- A 95% confidence interval means the area under the curve between two bounds is 0.95.
- This region is centered around the sample mean $\bar{X}$ and spans:

$$\bar{X} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

- So, 95% of the values (sample means from repeated sampling) fall in this region.
- We are saying: “We are 95% confident that the true mean μ lies within this range.”

$$\phi(x)$$

95% Confidence Interval in Standard Normal Distribution



Interpretation of 95% Confidence Interval Plot

- The plot shows the **standard normal distribution** $\mathcal{N}(0, 1)$, centered at 0.
- The shaded region represents the **central 95% of the probability mass**, between the critical values:

$$-1.96 \leq Z \leq 1.96$$

- The vertical dashed red lines at $z = -1.96$ and $z = 1.96$ are called **critical values**. These mark the boundaries beyond which lie the outer 5% of the distribution (i.e., 2.5% in each tail).
- The area under the curve between these lines is:

$$P(-1.96 < Z < 1.96) = 0.95$$

- This means that if we were to take many samples, 95% of their sample means would fall within this interval,
- The vertical black line at $z = 0$ indicates the **population mean**.

Confidence Interval Formula

For a population mean:

$$CI = \bar{x} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Where:

- $Z_{\alpha/2}$ = Critical value from the standard normal distribution.
- σ = Population standard deviation.
- n = Sample size.
- $\hat{p}$ = Sample proportion.

Interpreting Confidence Intervals

- A confidence interval gives a range of values within which the population parameter is expected to fall.
- Example: "We are 95% confident that the population mean lies between 45 and 55."
- **Common Misconception:** It does not mean the probability of containing the true parameter is 95% (the parameter is fixed, but the interval varies in repeated sampling).

Example: Point and Interval Estimates

Scenario: Estimating Average Height

- A **random sample of 100 students** has an average height of 170 cm with a standard deviation of 10 cm.
- **Point Estimate:** The best estimate of the population mean is $\bar{x} = 170$.
- **95% Confidence Interval:**

$$170 \pm 1.96 \times \frac{10}{\sqrt{100}} = 170 \pm 1.96$$

$$CI = (168.04, 171.96)$$

- **Conclusion:** The population mean height is likely between 168.04 cm and 171.96 cm.

Comparison: Point vs Interval Estimates

Point Estimate	Interval Estimate
Single numerical value	Range of values
No measure of uncertainty	Provides a confidence level
Easy to compute	Requires statistical formulas
Less reliable	More informative

Practical Applications of Estimation

- **Medical Studies:** Estimating the effect of a new drug.
- **Election Polls:** Predicting voter preferences with confidence intervals.
- **Quality Control:** Checking if manufactured products meet specifications.
- **Economics:** Estimating GDP growth rates with margin of error.

LAB: 3a

Hypothesis Testing

What is Hypothesis Testing?

- Hypothesis testing is a **statistical method** used to make decisions based on sample data.
- It helps determine whether there is **enough evidence** to reject a given assumption about a population.
- Used in **research, business, medicine**, and many other fields.
- **Hypothesis** is a claim or statement about a population parameter.

Real-World Scenario: Phone Battery Life

- A phone company claims its new model lasts **20 hours** on average.
- You, a tech reviewer, believe it lasts **less than 20 hours**.
- You test a sample of **30 phones** and record their battery life.
- Our conclusion is based on this sample.

Step 1: State the Hypotheses

- **Null Hypothesis (H_0):** Mean battery life is 20 hours
 $H_0 : \mu = 20$
- **Alternative Hypothesis (H_1):** Mean battery life is less than 20 hours
 $H_1 : \mu < 20$

- We do not know if H_0 is true or H_1 is true.
- We use statistics to decide whether to reject H_0 or fail to reject H_0 (Accept H_1).

Steps 2 & 3: Collect Data & Choose the Test

- **Sample size** (n) = 30
- **Sample mean** ($\bar{x}$) = 18.7 hours
- **Sample std. dev.** (s) = 2 hours
- Use a **one-sample t-test**
- Significance level $\alpha = 0.05$

Step 4: Calculate the Test Statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{18.7 - 20}{2/\sqrt{30}} \approx -3.56$$

- Degrees of freedom = 29
- Critical value (one-tailed, $\alpha = 0.05$) ≈ -1.699
- Since $t = -3.56 < -1.699$, **reject** the null hypothesis

What is a Test Statistic?

- A test statistic measures how far the **sample mean** ($\bar{x}$) is from the **hypothesized population mean** (μ_0).

- For a one-sample t-test:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

- It tells us how many **standard errors** the observed mean is away from the claimed mean.

Why is Standard Error Used?

- We are not testing an individual data point, but a **sample mean**.
- The sample mean has its own variability — measured by the **standard error**.
- Standard error (SE) is:

$$SE = \frac{s}{\sqrt{n}}$$

- It reflects how much sample means vary from the true mean.

Role of the Central Limit Theorem (CLT)

- CLT says: **Sampling distribution of the sample mean** is approximately normal if n is large enough.
- Properties:
 - Mean = μ
 - SD = $\sigma/\sqrt{n}$ (standard error)
- If σ is unknown, we use sample standard deviation s and apply the **t-distribution**.

Summary

- **Test statistic** = how many SEs the sample mean is from μ_0
- **Standard error** = variability of the sample mean
- **CLT** = Justifies using the t-distribution for $\bar{x}$

Z-score vs Test Statistic

Feature	Z-score	Test Statistic (t or z)
Measures	Distance from mean in standard units	Distance from hypothesized value in standard error (SE)
Formula	$z = \frac{x - \mu}{\sigma}$	$\text{stat} = \frac{\bar{x} - \mu_0}{SE}$
Interpretation	"How extreme is this value?"	"How extreme is this sample result?"
Standardized?	Yes	Yes
Centered at zero?	Yes	Yes (under null hypothesis)

Conclusion Based on Sample Mean

- **Claimed mean:** $\mu_0 = 20$ hours
- **Sample mean:** $\bar{x} = 18.7$ hours
- Since $18.7 < 20$, the sample suggests battery life is lower than claimed.
- This supports our suspicion that the company's claim is inaccurate.

Conclusion Based on Critical Value

- **Test statistic:** $t = -3.56$
- **Critical value (left-tailed, $\alpha = 0.05$):** $t_{crit} = -1.699$
- Since $-3.56 < -1.699$, the test statistic lies in the **rejection region**.
- **Conclusion:** Reject the null hypothesis (H_0).

Conclusion Based on Significance Level

- **Significance level:** $\alpha = 0.05$
 - This means we allow a 5% chance of wrongly rejecting H_0 .
 - Our test statistic is far into the rejection zone.
 - **Conclusion:** The evidence is strong enough to reject H_0 .
-
- We do not Know if H_0 is true or H_1 is true.
 - We collect a sample, and compute test statistic.
 - Measure its distance From Hypothesized value.
 - Decide whether to Reject or Fail to Reject H_0 .

Conclusion Based on p-value

- **Calculated p-value:** ≈ 0.0006
- **Compare with α :** $0.0006 < 0.05$
- p-value is very small $\rightarrow$ the result is **statistically significant**.
- **Conclusion:** Reject the null hypothesis. The battery life is likely less than 20 hours.

Step 5: Conclusion

- There is strong evidence that the average battery life is **less than 20 hours**.
- The company's claim does not hold based on our test.

Key Takeaway

- Hypothesis testing helps us decide if a claim is likely to be true.
- We use sample data and probability to make informed decisions.
- In this case, we statistically showed the phone's battery life is likely less than claimed.

Null Hypothesis (H_0)

- The **Null Hypothesis** (H_0) is the assumption that there is **no effect** or **no significant difference**.
- It represents the **status quo** or the assumption we begin with.
- **Example:**
 - H_0 : The new drug has **no effect** on blood pressure.
 - H_0 : There is **no difference** in test scores between two teaching methods.

Alternative Hypothesis (H_A or H_1)

- The **Alternative Hypothesis** (H_A) is the claim that contradicts the null hypothesis.
- It suggests that there **is an effect** or a **significant difference**.
- **Example:**
 - H_A : The new drug **reduces blood pressure**.
 - H_A : One teaching method is **more effective** than the other.

Why Focus on the Null Hypothesis?

- Hypothesis testing is designed to **disprove** rather than prove.
 - We assume H_0 is true unless there is **strong evidence** against it.
 - This approach helps in:
 - Avoiding **bias** in decision-making.
 - Ensuring a **scientific approach** to testing claims.
 - Reducing the chances of **false positives**.
 - If H_0 is rejected, we conclude that the evidence supports H_A .
- Similar to Judiciary Principle, Innocent until proven guilty (H_0).
 - If evidence is not sufficient, accused is declared innocent.
 - If sufficient evidence is provided, accused is declared guilty.

Errors in Hypothesis Testing

Two types of errors:

- **Type I Error** : Rejecting H_0 when it is actually true (**false positive**).
- **Type II Error** : Failing to reject H_0 when it is actually false (**false negative**).

Example:

- Type I Error: Concluding a drug works when it actually does not.
- Type II Error: Concluding a drug does not work when it actually does.

How to Choose Between Left, Right, and Two-Tailed Tests

- Choose the tail based on what the problem is testing:

Wording in Problem	Test Type	Alternative Hypothesis
"Greater than", "Increased", "More"	Right-tailed	$H_1 : \mu > \mu_0$
"Less than", "Reduced", "Dropped"	Left-tailed	$H_1 : \mu < \mu_0$
"Different", "Not equal", "Changed"	Two-tailed	$H_1 : \mu \neq \mu_0$

- Look for keywords in the question to decide the direction of the test.
- Use two-tailed when you're checking for any difference, not direction.

Hypothesis Testing

- Significance Level.

What is Significance Level (α)?

- The **significance level** (α) is the probability of making a **Type I error** (rejecting a true null hypothesis).
- It represents the **threshold** for determining whether an observed effect is statistically significant.
- **Common values of α :**
 - 0.05 (5%) - Most commonly used
 - 0.01 (1%) - More strict
 - 0.10 (10%) - Less strict

Example: If $\alpha = 0.05$, we accept a **5% chance** of incorrectly rejecting H_0 .

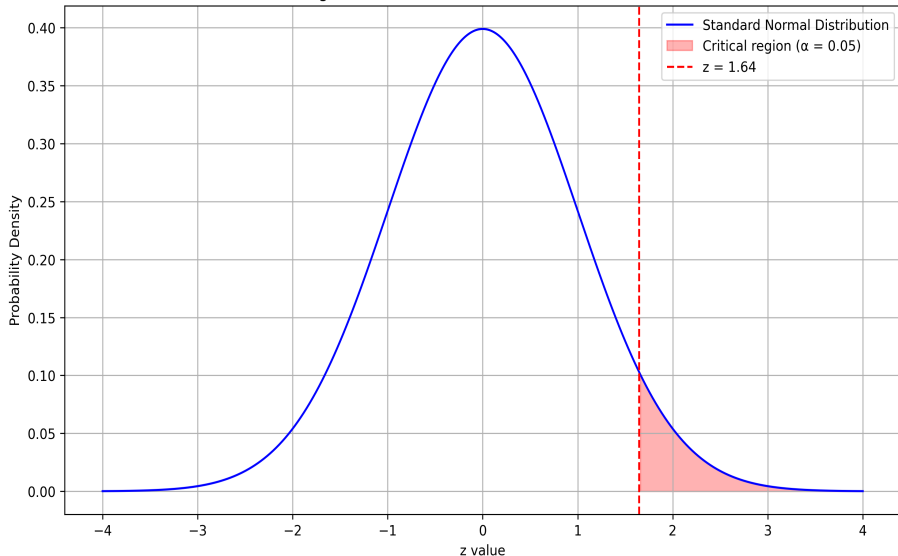
Type I and Type II Errors

Decision	Null Hypothesis True	Null Hypothesis False
Fail to Reject H_0	Correct Decision	Type II Error (β)
Reject H_0	Type I Error (α)	Correct Decision

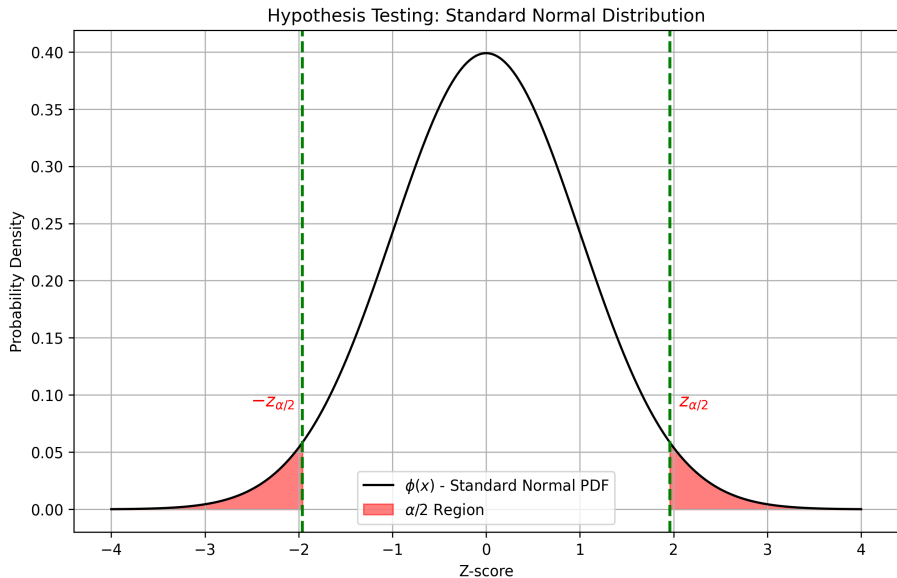
Table: Outcomes in Hypothesis Testing

α for right tailed test

Right-Tailed Test: Standard Normal Distribution

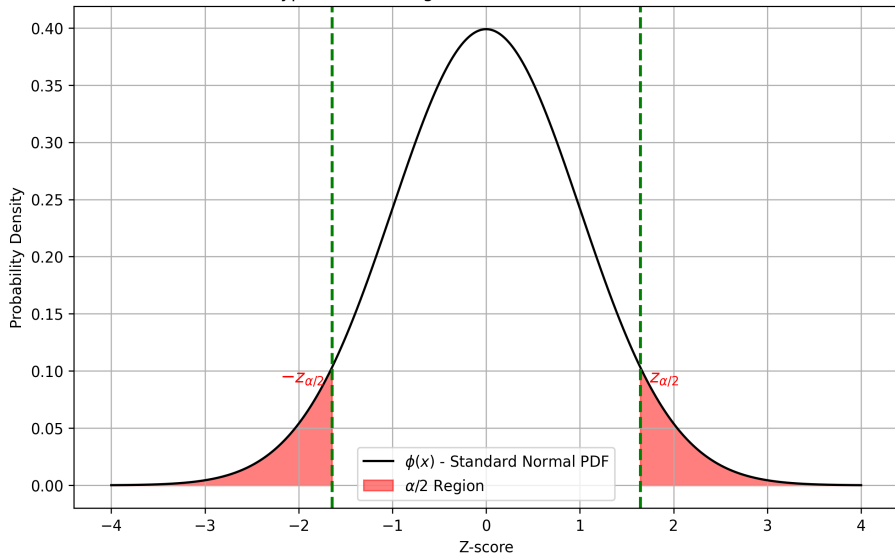


$\alpha/2$ region for $\alpha = 0.05$ for two-tailed test



$\alpha/2$ region for $\alpha = 0.1$ for two tailed test

Hypothesis Testing: Standard Normal Distribution



α and Critical Value in Hypothesis Testing

Significance Level (α):

- Probability of rejecting the null hypothesis when it is true.
- Represents the risk of a Type I error.
- Common values: $\alpha = 0.05, 0.01, 0.10$

Critical Value:

- Threshold that defines the boundary of the rejection region.
- Depends on:
 - α
 - Type of test (one-tailed or two-tailed)
 - Distribution used (e.g., standard normal, t -distribution)

Relationship Between α and Critical Value

One-tailed Z-test (right-tailed):

$$P(Z > z_{\alpha}) = \alpha$$

Left-tailed:

$$P(Z < z_{1-\alpha}) = \alpha$$

Two-tailed:

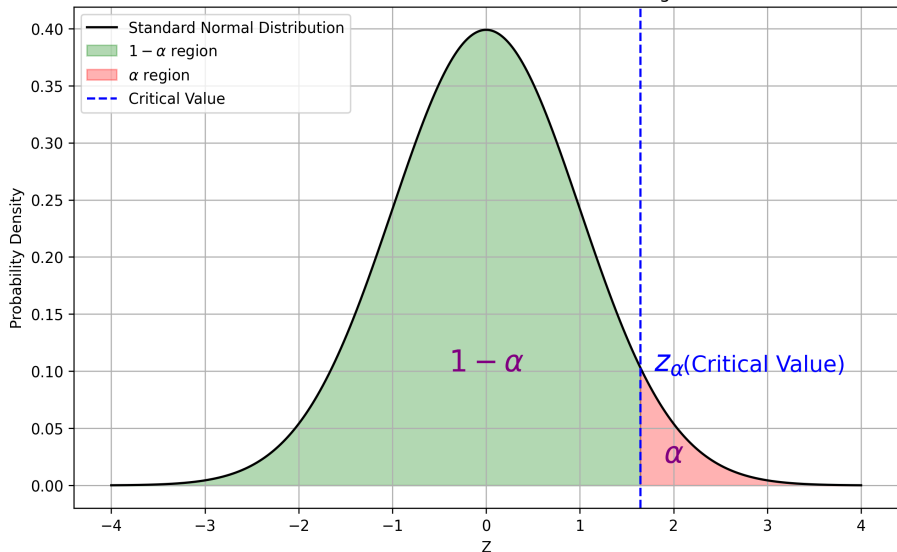
$$P(|Z| > z_{\alpha/2}) = \alpha$$

Example: For $\alpha = 0.05$

- One-tailed: $z_{\alpha} = z_{0.95} \approx 1.645$
- Two-tailed: $z_{\alpha/2} = z_{0.975} \approx \pm 1.96$

Rejection and Acceptance Region

Standard Normal Distribution with $1 - \alpha$ Region



Can Significance Level Be Zero?

No — the significance level (α) cannot be zero.

- α is the probability of making a **Type I error**:

Rejecting H_0 when H_0 is true

- Common values: $\alpha = 0.05, 0.01, 0.10$
- If $\alpha = 0$, you're saying:
 - “I will never reject the null hypothesis.”
 - No matter how strong the evidence is.

Why $\alpha = 0$ Makes No Sense

- If $\alpha = 0$, then:

Reject H_0 only if p-value < 0

- But **p-values can never be negative!**
- So, you will **never reject** H_0 — even if it's false.
- This leads to a **100% chance of Type II error:**

Failing to reject H_0 when it is false

Conclusion: Always set $\alpha > 0$ (e.g., 0.01, 0.05). A balance is needed between Type I and Type II errors.

Hypothesis Testing

- ✓ Significance Level.
- Test Statistic.

What is a Test Statistic?

Definition:

- A test statistic is a value computed from sample data
- It measures how far the observed result is from what is expected under the null hypothesis
- It helps us decide whether to reject the null hypothesis

General Formula:

$$\text{Test Statistic} = \frac{\text{Observed value} - \text{Hypothesized value}}{\text{Standard error}}$$

Example (Z-test):

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} \quad (\text{for known population standard deviation})$$

Test Statistic: Numerical Example

Given:

- Sample mean: $\bar{x} = 102$
- Hypothesized mean: $\mu_0 = 100$
- Population standard deviation: $\sigma = 10$
- Sample size: $n = 25$

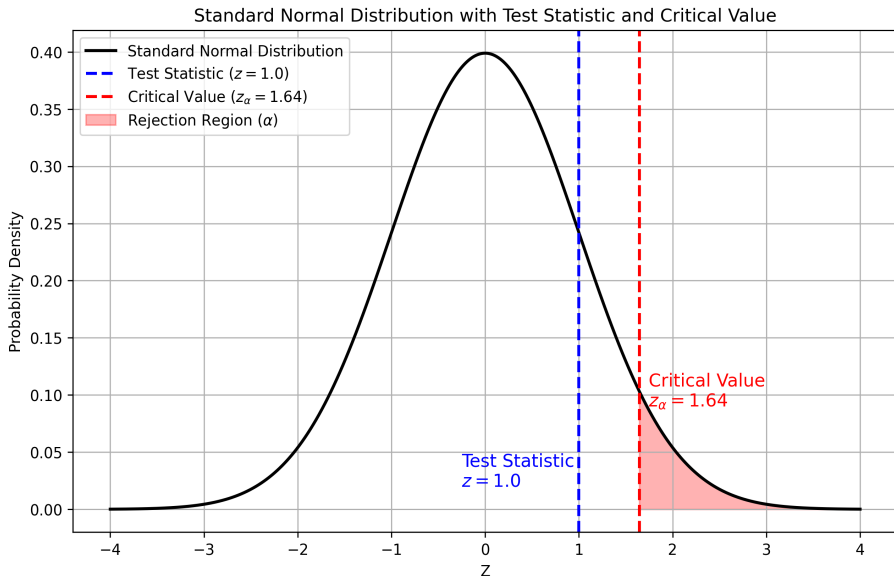
Z-test formula:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{102 - 100}{10/\sqrt{25}} = \frac{2}{2} = 1.0$$

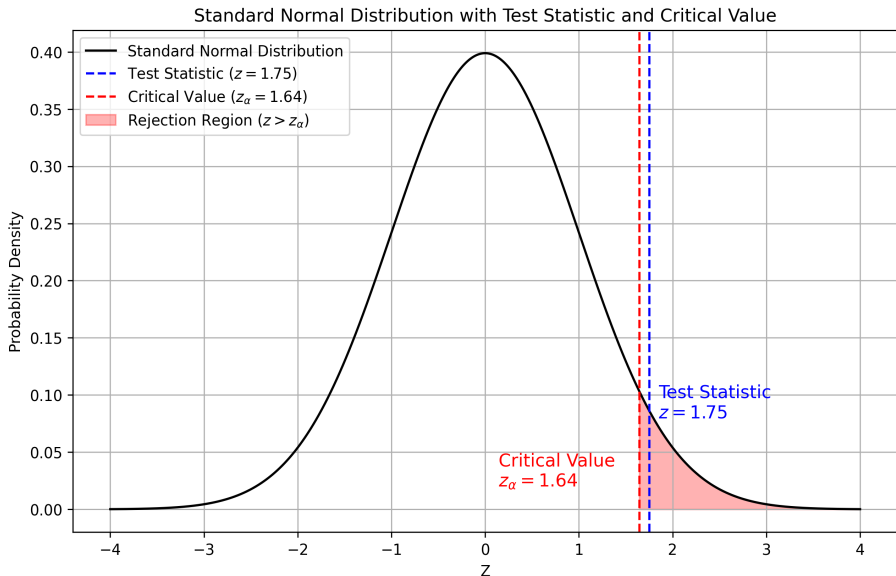
Interpretation:

- The test statistic $z = 1.0$ indicates the sample mean is 1 standard error above the hypothesized mean.
- Compare this value to the critical value to make a decision.

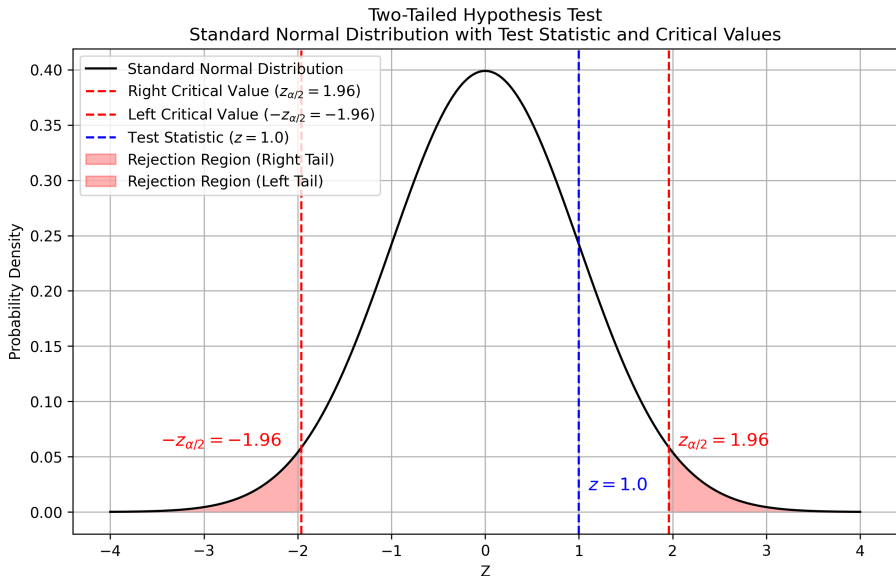
Test Statistic and Critical Value



Test Statistic and Critical Value



Test Statistic and Critical Value



Test Statistic and Critical Value

Both the test statistic and the critical value are points on the x-axis of the sampling distribution.

Hypothesis Testing

- ✓ Significance Level.
- ✓ Test Statistic.
- p-value.

What is the p-value?

- The **p-value** is the probability of observing the test results, or something more extreme, assuming H_0 is true.
- It measures the **strength of evidence** against H_0 .
- **Key idea:**
 - A **small p-value** suggests strong evidence against H_0 .
 - A **large p-value** suggests weak or no evidence against H_0 .

Example: If $p = 0.02$, there is only a 2% chance of observing the data assuming H_0 is true.

How is the p-value Determined?

- The p-value is computed using a **test statistic**, which depends on the type of hypothesis test.
- Steps to compute the p-value:
 - ① Choose a hypothesis test (**t-test, z-test, chi-square test, etc.**).
 - ② Compute the **test statistic** based on sample data.
 - ③ Find the probability (p-value) using the statistical distribution.

Example: In a two-tailed test using a z-score:

- If $z = 2.1$, we find $p = 0.035$ from the normal distribution table.

How to Interpret the p-value?

General Interpretation:

- If $p < \alpha$, reject H_0 (evidence suggests a significant effect).
- If $p \geq \alpha$, fail to reject H_0 (not enough evidence to support H_A).

Example for $\alpha = 0.05$:

- If $p = 0.02$, reject H_0 (strong evidence against H_0).
- If $p = 0.08$, fail to reject H_0 (not enough evidence).

Significance level and p-Value

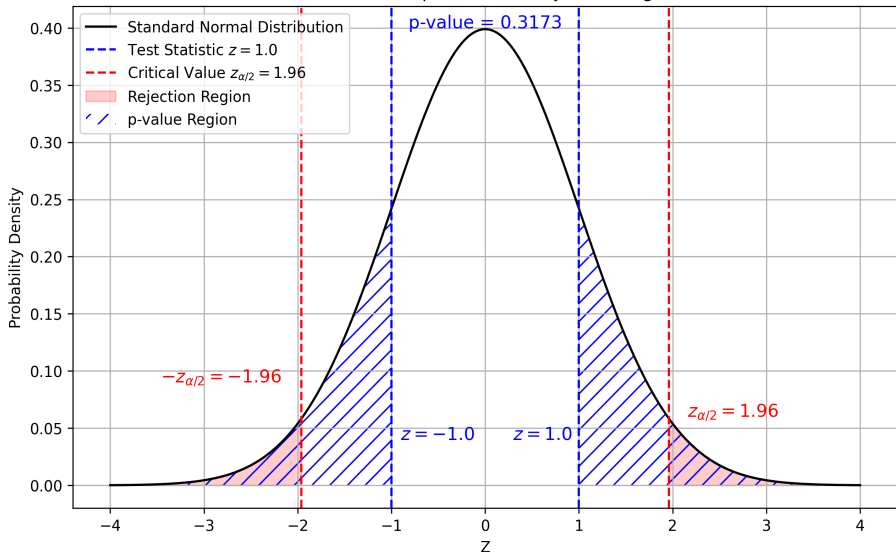
Both the significance level (α) and the p-value represent areas under the sampling distribution curve.

Conclusion

- The **significance level** (α) defines the threshold for statistical significance.
- The **p-value** measures the probability of obtaining the observed data under H_0 .
- A **small p-value** ($p < \alpha$) provides strong evidence against H_0 .
- If $p \geq \alpha$, we **fail to reject** H_0 .

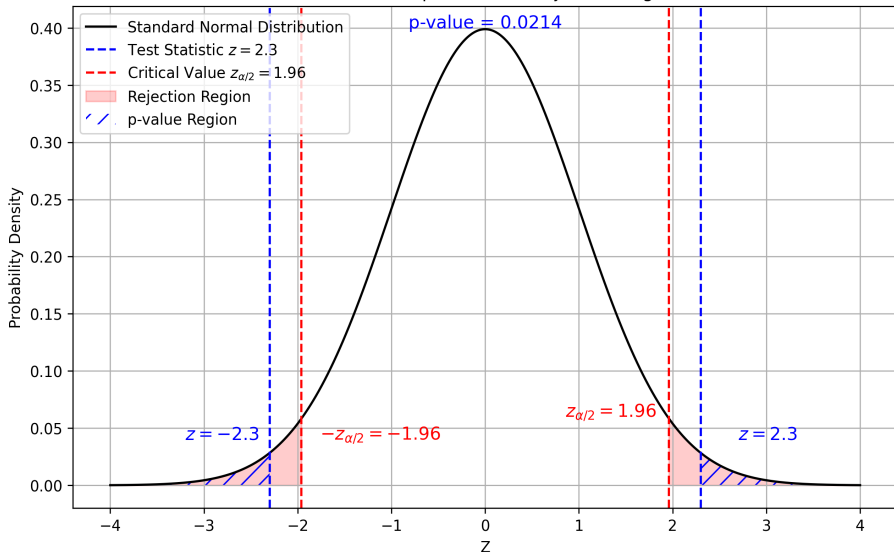
Test Statistic and Critical Value, p-value and $\alpha = 0.05$

Two-Tailed Test: p-value and Rejection Regions



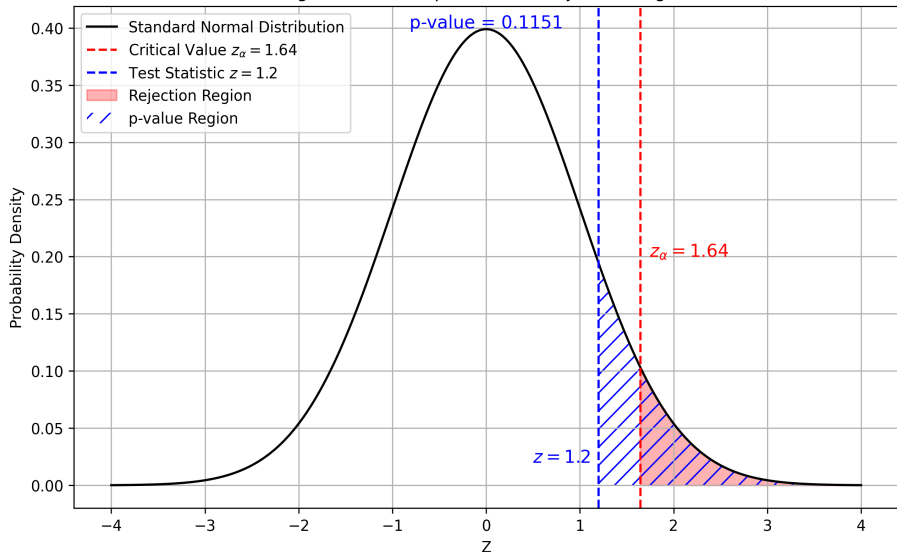
Test Statistic and Critical Value, p-value and $\alpha = 0.05$

Two-Tailed Test: p-value and Rejection Regions



Test Statistic and Critical Value, p-value and $\alpha = 0.05$

Right-Tailed Test: p-value and Rejection Region



critical values for $\alpha = 0.05$ for two tailed and right tailed test

- Observe critical values for the two tests.
- Comment on them.

Hypothesis Testing

- ✓ Significance Level.
- ✓ Test Statistic.
- ✓ p-value.
- Steps in hypothesis testing.

Steps in Hypothesis Testing

- 1 **State the hypotheses** (H_0 and H_A).
- 2 **Choose a significance level** (α , commonly 0.05).
- 3 **Select a test statistic** (e.g., t-test, z-test, chi-square test).
- 4 **Compute the test statistic and p-value.**
- 5 **Make a decision:**
 - If $p < \alpha$, **reject** H_0 .
 - If $p \geq \alpha$, **fail to reject** H_0 .

Example: Testing a New Diet Plan

Scenario: A researcher claims a new diet reduces weight.

Step 1: State Hypotheses

- H_0 : The diet has no effect on weight loss.
- H_A : The diet leads to weight loss.

Step 2: Significance Level

- Choose $\alpha = 0.05$.

Step 3: Compute Test Statistic and p-value

- If $p = 0.03 < 0.05$, **reject** H_0 .

Conclusion: The diet plan is statistically effective!

Test Statistic vs Critical Value and p-value vs Significance Level

Concept Pair	What They Represent	Location in Distribution
Test Statistic vs Critical Value	They are both numerical values used for comparison in hypothesis testing	Points on the x-axis of the sampling distribution
p-value vs Significance Level (α)	They both represent probabilities or areas used to make decisions	Areas under the sampling distribution curve

How These Work Together: Decision Rules

Comparison	What It Means	Decision Rule
Test Statistic vs Critical Value	Compares the calculated result to the threshold under H_0	Reject H_0 if test statistic falls in the rejection region (beyond critical value)
p-value vs Significance Level (α)	Compares observed probability to allowed error risk	Reject H_0 if p-value $< \alpha$

Type I, Type II Errors and power of a test

Errors in Hypothesis Testing and Power

- **Type I Error (α):** Rejecting H_0 when it is actually true $\rightarrow$ "False positive"
- **Type II Error (β):** Failing to reject H_0 when H_1 is actually true $\rightarrow$ "False negative" (missed detecting the truth of H_1)
- **Power of the Test ($1 - \beta$):** Correctly rejecting H_0 when H_1 is true $\rightarrow$ "Detecting the effect when H_1 is true"

- Similar to Judiciary Principle, Innocent until proven guilty (H_0).
- If evidence is not sufficient, accused is declared innocent.
- If sufficient evidence is provided, accused is declared guilty.
- Enough evidence : Accused is declared guilty and sentenced, but accused is actually Innocent (**Type - 1 Error**).
- Not Enough evidence : Accused is set free, but accused is actually guilty (**Type - 2 error**).
- Enough evidence : Accused is declared guilty and sentenced, and accused is actually Guilty (**Power of the test**).

Type I and Type II Errors

Decision	Null Hypothesis True	Null Hypothesis False
Fail to Reject H_0	Correct Decision	Type II Error (β)
Reject H_0	Type I Error (α)	Correct Decision

Table: Outcomes in Hypothesis Testing

Effect of Varying α on Errors and Power

Metric	When α Increases	When α Decreases
Type I Error (α)	↑ Increases	↓ Decreases
Type II Error (β)	↓ Decreases	↑ Increases
Power ($1 - \beta$)	↑ Increases	↓ Decreases

Probabilistic Interpretation of Errors and Power

- If H_0 is true, there is an $\alpha\%$ chance of making a **Type I error** (rejecting a true null).
- If H_1 is true, there is a $\beta\%$ chance of making a **Type II error** (failing to reject a false null).
- The **power of the test** is $1 - \beta$, the probability of correctly rejecting H_0 when H_1 is true.

Trade-off Between Type I and Type II Errors

- Reducing the probability of one type of error usually increases the probability of the other.
- **Lowering** the significance level (α) reduces Type I Errors but increases Type II Errors.
- **Increasing** statistical power ($1 - \beta$) reduces Type II Errors but may increase Type I Errors.

Example:

- A strict court system may convict more innocent people (higher Type I Error).
- A lenient court system may acquit more guilty people (higher Type II Error).

Scenario 1: Courtroom Decision

Hypothesis:

- H_0 : The defendant is innocent.
- H_A : The defendant is guilty.

Errors:

- **Type I Error:** Convicting an innocent person.
- **Type II Error:** Acquitting a guilty person.

Implications:

- A strict legal system may reduce Type II Errors but increase Type I Errors.
- A lenient legal system reduces Type I Errors but increases Type II Errors.

Scenario 2: Medical Testing

Hypothesis:

- H_0 : The patient does not have the disease.
- H_A : The patient has the disease.

Errors:

- **Type I Error:** Diagnosing a healthy person as sick (false positive).
- **Type II Error:** Failing to detect the disease in a sick person (false negative).

Implications:

- A test with high sensitivity reduces Type II Errors but may increase Type I Errors.
- A test with high specificity reduces Type I Errors but may increase Type II Errors.

Scenario 3: Quality Control in Manufacturing

Hypothesis:

- H_0 : The product meets quality standards.
- H_A : The product is defective.

Errors:

- **Type I Error:** Rejecting a good product (false alarm).
- **Type II Error:** Accepting a defective product.

Implications:

- Stricter quality control reduces Type II Errors but may increase waste.
- Lenient quality control reduces waste but may increase faulty products reaching customers.

Sample Size

What is Sample Size?

- Sample size (n) is the number of observations in a study.
- It determines the precision and reliability of the results.
- A larger sample size reduces random variability and increases confidence in findings.

Why Sample Size Matters?

Example: Medical Trial for a New Drug

- A pharmaceutical company wants to test if a new drug reduces blood pressure.
- They conduct two trials:
 - ① **Trial 1:** 10 patients, results are highly variable.
 - ② **Trial 2:** 1000 patients, results are more reliable.
- Conclusion: **A small sample may give misleading results, while a large sample increases reliability.**

Factors Affecting Sample Size

- **Population Variability** (σ): Higher variability requires a larger sample.
- **Confidence Level** ($1 - \alpha$): Higher confidence levels need more samples.
- **Margin of Error** (E): Smaller errors require larger samples.
- **Effect Size** (d): Detecting smaller differences requires more data.
- **Statistical Power** ($1 - \beta$): Higher power needs a larger sample.

Sample Size Formula (Mean)

For Estimating a Population Mean:

$$n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{E} \right)^2$$

Where:

- n = Required sample size
- $Z_{\alpha/2}$ = Critical value from the standard normal distribution
- σ = Population standard deviation
- E = Margin of error

Sample Size Formula (Proportions)

For Estimating a Population Proportion:

$$n = \frac{Z_{\alpha/2}^2 \cdot p \cdot (1 - p)}{E^2}$$

Where:

- p = Estimated proportion (e.g., proportion of success/failure)
- Other variables are the same as before.

Effect of Sample Size on Confidence Interval

- The confidence interval (CI) for a mean is given by:

$$\bar{x} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

- As n increases:
 - $\frac{\sigma}{\sqrt{n}}$ (Standard Error) decreases.
 - The confidence interval becomes **narrower**.
 - Results become more precise and reliable.

Effect of Sample Size on Margin of Error

- The margin of error (E) is:

$$E = Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

- A larger sample size decreases E , meaning:
 - Estimates are closer to the true population value.
 - We gain more confidence in our conclusions.

Step-by-Step Example

Example: Estimating the Average Weight of Students

- Desired confidence level: 95% ($Z_{\alpha/2} = 1.96$)
- Population standard deviation: $\sigma = 10$ kg
- Margin of error: $E = 2$ kg

Calculation:

$$n = \left(\frac{1.96 \times 10}{2} \right)^2 = 96$$

Conclusion: We need at least **96 students** for reliable results.

Sample Size and Statistical Power

- **Power of a test** ($1 - \beta$) is the probability of detecting a true effect.
- Increasing sample size increases power, reducing the risk of Type II errors.
- A common standard is **80% power** ($\beta = 0.2$).

Practical Applications of Sample Size Calculation

- **Medical Research:** Ensuring enough patients are tested for reliable results.
- **Marketing Surveys:** Collecting sufficient responses for accurate consumer insights.
- **Quality Control:** Ensuring sampled products represent the entire production batch.
- **Election Polls:** Determining how many people to survey for accurate predictions.

Testing

Introduction to Testing

- Statistical tests are used to make inferences about a population based on sample data.
- Different tests are applied depending on whether we are analyzing means, variances, or proportions.
- These tests help us determine if observed differences are due to real effects or random chance.
- They are widely used in fields such as science, business, healthcare, and engineering for decision-making.

Steps in Hypothesis Testing

1. Formulate the Hypotheses

- Null Hypothesis (H_0): Assumes no effect or no difference.
- Alternative Hypothesis (H_a): Represents what we seek to prove.

2. Choose the Significance Level (α)

- Common values are 0.05 (5%) or 0.01 (1%).
- Defines the probability of rejecting H_0 when it is actually true.

3. Select the Appropriate Test Statistic

- Based on sample size, data type, and known parameters.
- Examples: t-test, Z-test, chi-square test, F-test.

Steps in Hypothesis Testing (Contd.)

4. Compute the Test Statistic

- Apply the relevant formula for the selected test.
- Use sample data to compute the test value.

5. Determine the Critical Value or P-Value

- Critical value method: Compare the test statistic to a threshold.
- P-value method: Check if p-value is less than significance level (α).

6. Make a Decision

- If test statistic exceeds critical value (or $p\text{-value} < \alpha$), reject H_0 .
- Otherwise, fail to reject H_0 (not enough evidence to support H_a).

Tests for Mean

Why are tests for mean needed?

- To compare the average (mean) of a sample with a known value or another sample.
- Helps determine if differences in means are statistically significant or due to random variation.

Common tests:

- **t-test:** Used when the population variance is unknown and sample size is small.
- **Z-test:** Used when the population variance is known and sample size is large.

Tests for Variance

Why are tests for variance needed?

- To assess if the variability in a dataset matches a theoretical expectation or differs between two groups.
- Variance affects reliability and precision in data analysis.

Common tests:

- **Chi-square test:** Compares a sample variance to a known population variance.
- **F-test:** Compares the variances of two independent samples.

Tests for Proportion

Why are tests for proportion needed?

- To determine if a sample proportion differs significantly from a known population proportion.
- Used frequently in surveys, elections, and quality control.

Common test:

- **Z-test for proportion:** Used when the sample size is sufficiently large (typically $n \cdot p \geq 10$ and $n \cdot (1 - p) \geq 10$).

One Sample Z-Test

One Sample Z-Test

One-Sample Z-Test:

- Used to **compare** the **mean of a sample** to a **known population mean**.
- Applicable when population standard deviation (σ) is known.
- Assumes normality or large sample size (Central Limit Theorem).

Hypotheses Null Hypothesis (H_0): $\mu = \mu_0$ (Sample mean equals population mean)

Alternative Hypothesis (H_a): $\mu \neq \mu_0$ (Two-tailed) or $\mu > \mu_0$ or $\mu < \mu_0$ (One-tailed)

Test Statistic

Formula:

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

- $\bar{x}$ = sample mean
- μ_0 = population mean
- σ = population standard deviation
- n = sample size

Decision Rule

- Choose significance level (α , e.g., 0.05).
- Find critical value from Z-table.
- If $|Z| > Z_{critical}$, reject H_0 .

Example

Scenario:

- A factory claims that the **mean weight** of its sugar packs is **500g**.
- A **random sample of 30 packs** has a **sample mean of 495g**.
- The **population standard deviation is 10g**. Test the claim at $\alpha = 0.05$.

Solution Steps

Step 1: Define Hypotheses

$$H_0: \mu = 500$$

$$H_a: \mu \neq 500 \text{ (Two-tailed test)}$$

Step 2: Compute Z-Statistic

$$\begin{aligned} Z &= \frac{495 - 500}{10/\sqrt{30}} \\ &= \frac{-5}{10/5.477} \\ &= \frac{-5}{1.826} \end{aligned}$$

Solution Steps (cont.)

Step 3: Compare with Critical Value

For $\alpha = 0.05$ (two-tailed), critical value = ± 1.96 from Z-table.

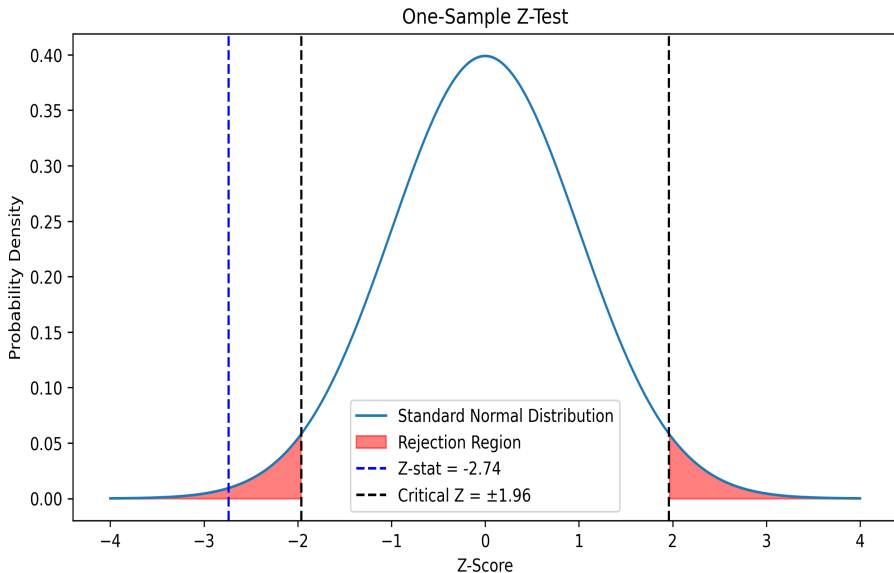
Since $-2.74 < -1.96$, we reject H_0 .

Conclusion

Since $Z = -2.74$ falls in the rejection region, we reject H_0 .

There is significant evidence that the mean weight is not 500g.

One Sample Z-Test

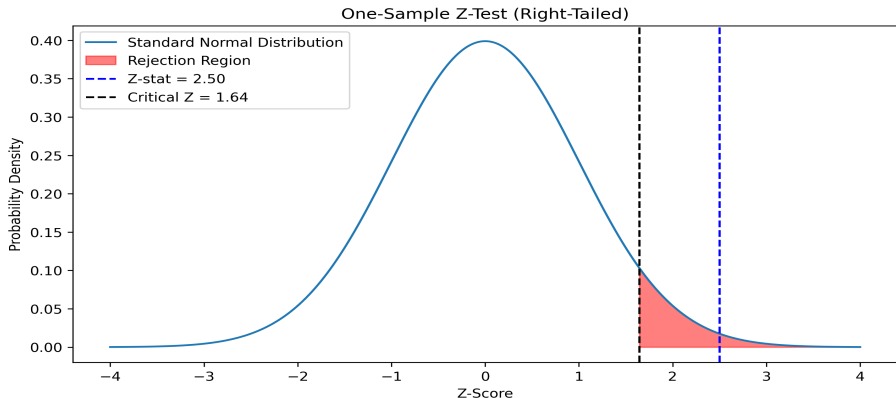


One Sample Z-Test

- A factory claims its perfume bottles have a **mean volume of 150cc**.
- Population standard deviation is **2cc**; sample size is **100 bottles**.
- The sample mean volume is **150.5cc**. Test at a **95% confidence level** whether the mean has **increased from 150cc**.

One Sample Z-Test

- A factory claims its perfume bottles have a **mean volume of 150cc**.
- Population standard deviation is **2cc**; sample size is **100 bottles**.
- The sample mean volume is **150.5cc**. Test at a **95% confidence level** whether the mean has **increased from 150cc**.



One Sample T-Test

One-Sample T-Test

One-Sample T-Test:

- Used to compare the mean of a sample to a known population mean when the population standard deviation is unknown.
- Assumes data is approximately normally distributed.
- Applicable for small sample sizes (typically $n < 30$).

Hypotheses Null Hypothesis (H_0): $\mu = \mu_0$ (Sample mean equals population mean)

Alternative Hypothesis (H_a): $\mu \neq \mu_0$ (Two-tailed) or $\mu > \mu_0$ or $\mu < \mu_0$ (One-tailed)

Test Statistic

Formula:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

- $\bar{x}$ = sample mean
- μ_0 = population mean
- s = sample standard deviation
- n = sample size

Decision Rule

- Choose significance level (α , e.g., 0.05).
- Find critical value from t-distribution table based on $df = n - 1$.
- If $|t| > t_{critical}$, reject H_0 .

Example

Scenario:

- A factory claims the **mean volume** of its perfume bottles is **150cc**.
- The **population standard deviation is unknown**.
- A **sample of 25 bottles** has a **mean of 151cc** and a **sample standard deviation of 1.5cc**.
- Test at a **95% confidence level** whether the mean volume has **increased from 150cc**.

Solution Steps

Step 1: Define Hypotheses

$$H_0: \mu = 150$$

$$H_a: \mu > 150 \text{ (One-tailed test)}$$

Step 2: Compute T-Statistic

$$\begin{aligned} t &= \frac{151 - 150}{1.5/\sqrt{25}} \\ &= \frac{1}{0.3} \\ &= 3.33 \end{aligned}$$

Solution Steps (cont.)

Step 3: Compare with Critical Value

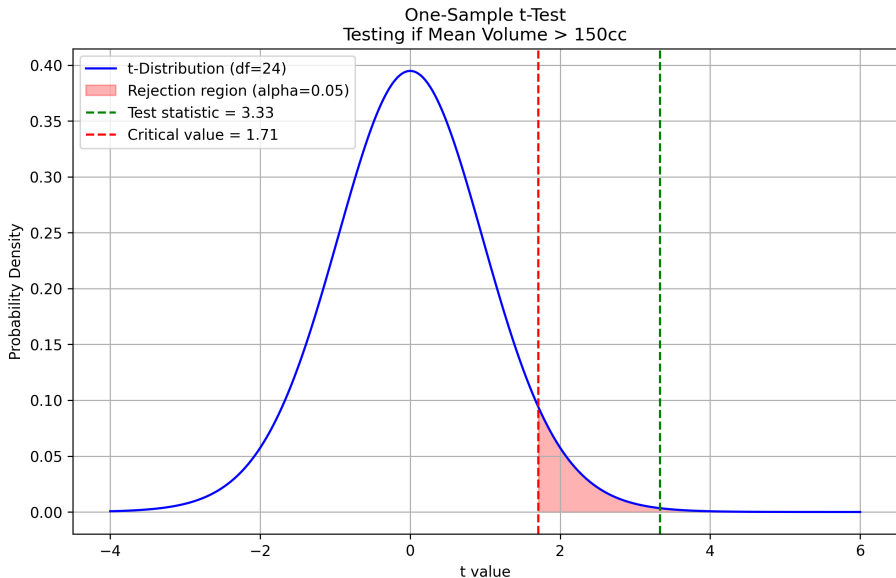
Degrees of freedom: $df = 25 - 1 = 24$

For $\alpha = 0.05$ (one-tailed), critical value from t-table = 1.711.

Since $3.33 > 1.711$, we reject H_0 .

Conclusion Since $t = 3.33$ falls in the rejection region, we reject H_0 . There is significant evidence that the mean volume has increased from 150cc.

One Sample T-Test



One Proportion Z-Test

One-Proportion Z-Test

One-Proportion Z-Test:

- Used to test if the proportion of a sample is significantly different from a hypothesized population proportion.
- Assumes a large enough sample size for normal approximation.
- Used for Categorical data.
- Often used in surveys, quality control, and medical studies.

Assumptions

- Data follows a binomial distribution.
- The sample size is large enough to approximate normality:

$$np_0 \geq 5, \quad n(1 - p_0) \geq 5$$

- The sample is randomly selected and independent.

Hypotheses

Null Hypothesis (H_0): $p = p_0$ (Sample proportion equals population proportion)

Alternative Hypothesis (H_a): $p \neq p_0$ (Two-tailed) or $p > p_0$ or $p < p_0$ (One-tailed)

Test Statistic

Formula:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

- $\hat{p}$ = sample proportion
- p_0 = population proportion
- n = sample size

Decision Rule

- Choose significance level (α , e.g., 0.05).
- Find critical value from standard normal (Z) table.
- If $|z| > z_{critical}$, reject H_0 .

Example

Scenario:

A company claims that 60% of customers are satisfied with their product. A random sample of 200 customers finds that 130 are satisfied. Test at a 95% confidence level whether the satisfaction rate is different from 60%.

Solution Steps Step 1: Define Hypotheses

$$H_0: p = 0.60$$

$$H_a: p \neq 0.60 \text{ (Two-tailed test)}$$

Solution Steps (cont.)

Step 2: Compute Test Statistic

$$\begin{aligned}\hat{p} &= \frac{130}{200} = 0.65 \\ z &= \frac{0.65 - 0.60}{\sqrt{\frac{0.60(1-0.60)}{200}}} \\ &= \frac{0.05}{0.0346} \\ &= 1.44\end{aligned}$$

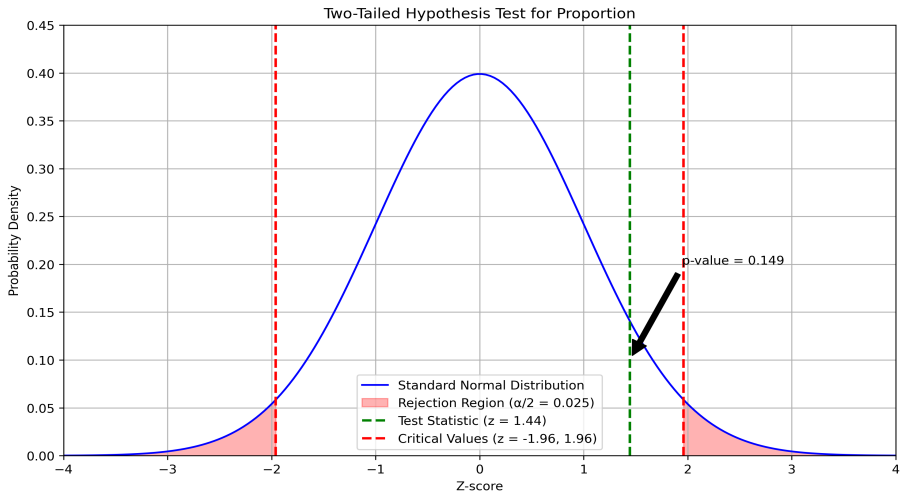
Step 3: Compare with Critical Value

For $\alpha = 0.05$ (two-tailed), critical value from Z-table = ± 1.96 .

Since $1.44 < 1.96$, we fail to reject H_0 .

There is no significant evidence that the satisfaction rate is different from 60%.

One Proportion Z-Test

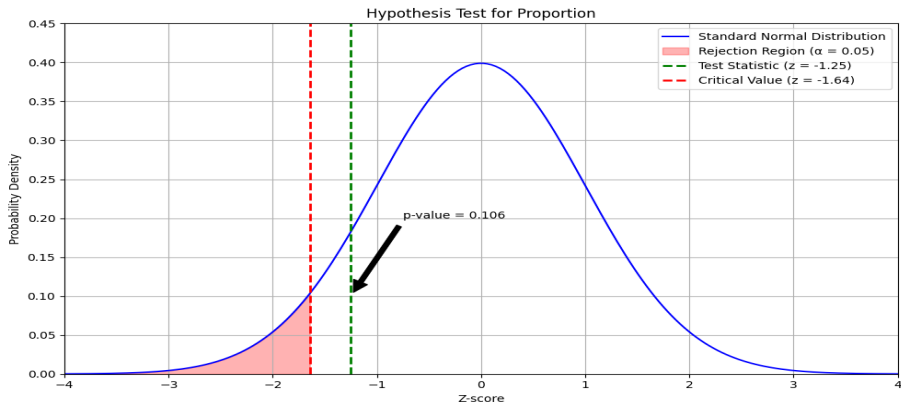


One Proportion Z-Test

- A factory claims that **80% of its products** pass quality inspection.
- A **random sample of 100 products** is taken, and **75 products pass**.
- Test at a **95% confidence level** whether the true proportion is **less than 80%**.

One Proportion Z-Test

- A factory claims that **80% of its products** pass quality inspection.
- A **random sample of 100 products** is taken, and **75 products pass**.
- Test at a **95% confidence level** whether the true proportion is **less than 80%**.



One Variance Test

One Variance test

One-Variance Test is used to determine whether the variance of a population is equal to a specified value.

Applications: Quality control, finance, manufacturing, etc.

Assumptions

- The population from which the sample is drawn follows a normal distribution.
- The sample is randomly selected.

Hypothesis Statements

Two-tailed Test:

- $H_0 : \sigma^2 = \sigma_0^2$ (The population variance is equal to a specified value.)
- $H_1 : \sigma^2 \neq \sigma_0^2$ (The population variance is different from the specified value.)

One-tailed Tests:

- Right-tailed: $H_1 : \sigma^2 > \sigma_0^2$
- Left-tailed: $H_1 : \sigma^2 < \sigma_0^2$

Test Statistic

The test statistic follows a chi-square distribution:

$$\chi^2 = \frac{(n - 1)s^2}{\sigma_0^2}$$

where:

- n = sample size
- s^2 = sample variance
- σ_0^2 = hypothesized population variance

Chi-Square Test for One Variance

Given:

- Sample size n
- Claimed variance σ_0^2
- Sample variance s^2
- Degrees of freedom $df = n - 1$

Test statistic:

$$\chi^2 = \frac{(n - 1)s^2}{\sigma_0^2}$$

- Degrees of freedom $df = n - 1$ because sample variance s^2 is calculated using sample mean. And sample mean has degrees of freedom $n - 1$.

How to Calculate the P-value

Once the χ^2 statistic is computed, the **p-value** depends on the type of test:

Test Type	P-value Calculation
Right-tailed	$p = P(\chi^2 \geq \text{test statistic})$
Left-tailed	$p = P(\chi^2 \leq \text{test statistic})$
Two-tailed	$p = 2 \times \min(P(\chi^2 \geq \text{test statistic}), P(\chi^2 \leq \text{test statistic}))$

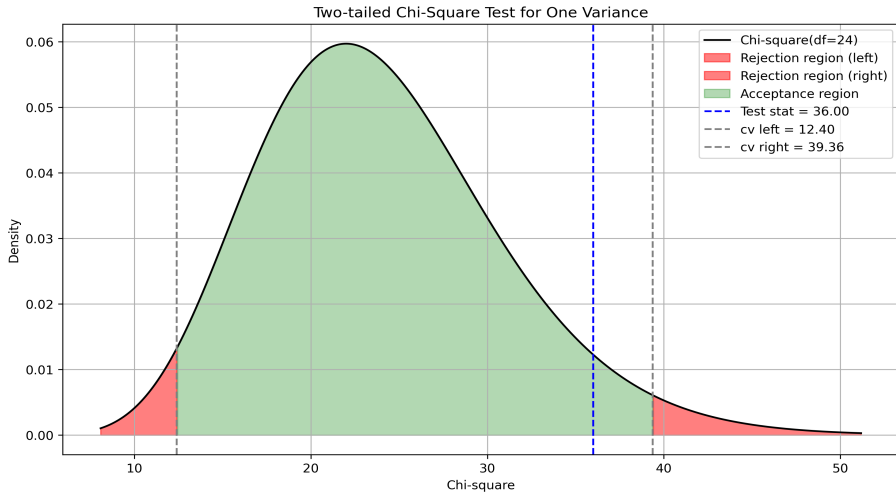
Interpretation of the P-value

- The p-value is the probability of observing a test statistic at least as extreme as the one observed under H_0 .
- It represents the area under the chi-square curve beyond (or before) the observed test statistic.
- Compare the p-value with the significance level α to decide whether to reject H_0 .

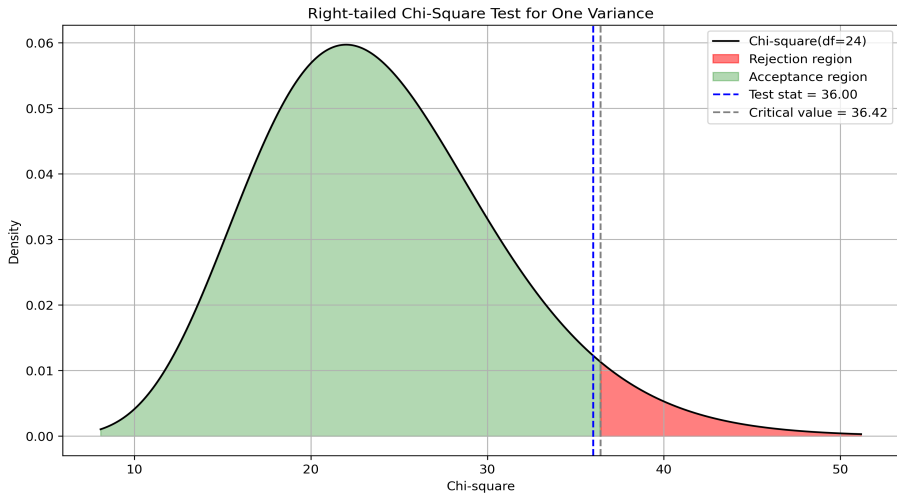
Decision Rule

- Compare the test statistic with the critical value from the chi-square distribution.
- Reject H_0 if:
 - Two-tailed: $\chi^2 < \chi_{\alpha/2, n-1}^2$ or $\chi^2 > \chi_{1-\alpha/2, n-1}^2$
 - Right-tailed: $\chi^2 > \chi_{1-\alpha, n-1}^2$
 - Left-tailed: $\chi^2 < \chi_{\alpha, n-1}^2$

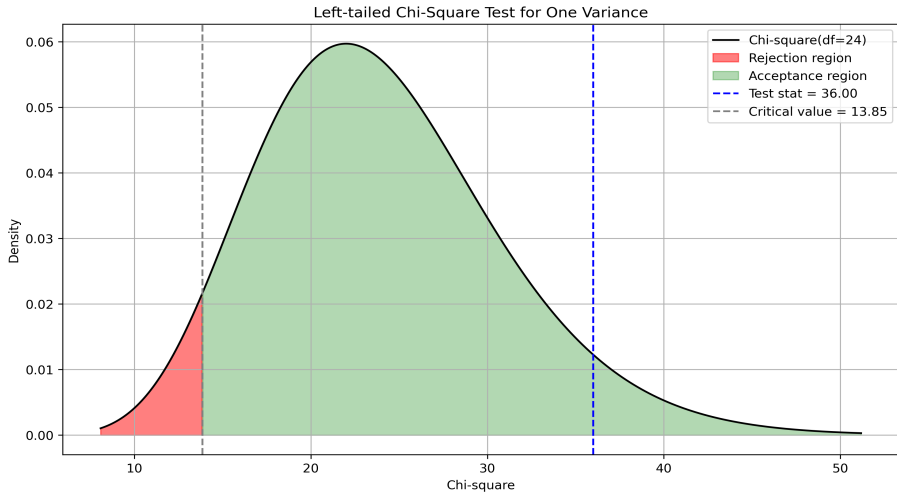
Two Tailed One Variance Test



Right Tailed One Variance Test



Left Tailed One Variance Test



Example

Problem: A factory claims that the variance of the weight of a product is 4 grams². A sample of 10 items has a variance of 6 grams². Test the claim at $\alpha = 0.05$.

Step 1: Define Hypothesis

- $H_0 : \sigma^2 = 4$
- $H_1 : \sigma^2 \neq 4$

Step 2: Compute Test Statistic

$$\chi^2 = \frac{(10 - 1) \times 6}{4} = \frac{54}{4} = 13.5$$

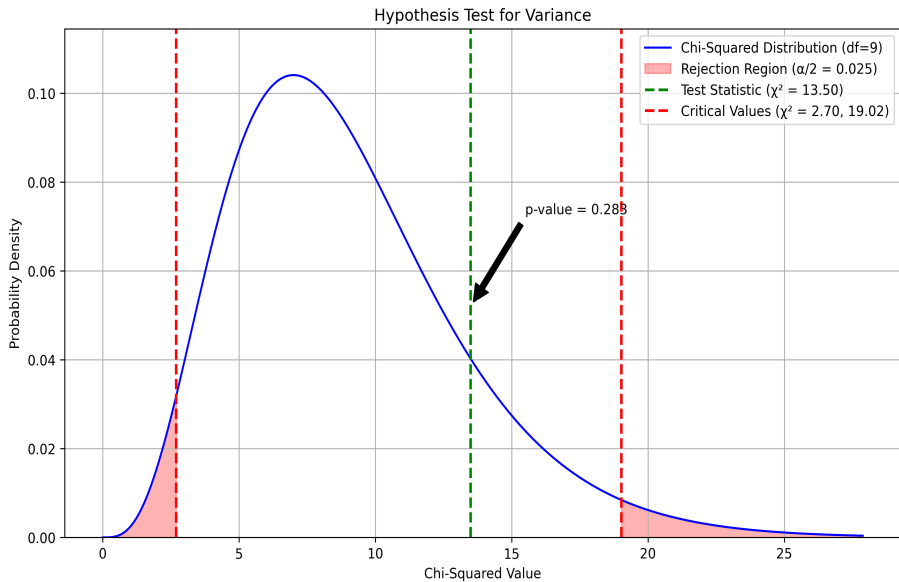
Step 3: Find Critical Values

- Using χ^2 table with $df = 9$, $\alpha/2 = 0.025$, critical values are:
- $\chi^2_{0.025,9} = 19.02$ and $\chi^2_{0.975,9} = 2.70$

Step 4: Decision

- Since $2.70 < 13.5 < 19.02$, we fail to reject H_0 .
- Conclusion: There is no significant evidence to say the variance is different from 4.

One variance Test

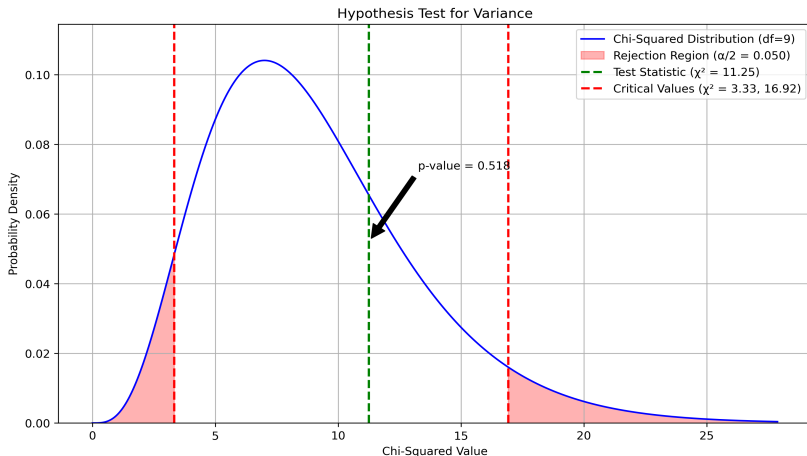


One variance Test

A manufacturer claims that the variance in the weight of their product is 4 grams². A sample of 10 products is tested, and the sample variance is found to be 5 grams². Test at a 90% confidence level whether the population variance is different from 4 grams².

One variance Test

A manufacturer claims that the variance in the weight of their product is 4 grams². A sample of 10 products is tested, and the sample variance is found to be 5 grams². Test at a 90% confidence level whether the population variance is different from 4 grams².



Two Sample Z-Test

Two-Sample Z-Test

Two-Sample Z-Test is used to determine whether there is a significant difference between the means of two independent populations.

Applications: Comparing average salaries, testing effectiveness of new drugs, analyzing customer behavior, etc.

Why Use a Two-Sample Z-Test?

- Used when comparing the means of two independent groups.
- Appropriate when population standard deviations (σ_1, σ_2) are known.
- Used for large sample sizes ($n_1, n_2 \geq 30$), ensuring normality by the Central Limit Theorem.

Assumptions

- The two samples are independent.
- The population standard deviations (σ_1 and σ_2) are known.
- Each sample is randomly selected.
- The samples are either large ($n_1, n_2 \geq 30$) or the population is normally distributed.

Hypothesis Statements

Two-tailed Test:

- $H_0 : \mu_1 = \mu_2$ (The population means are equal.)
- $H_1 : \mu_1 \neq \mu_2$ (The population means are different.)

One-tailed Tests:

- Right-tailed: $H_1 : \mu_1 > \mu_2$
- Left-tailed: $H_1 : \mu_1 < \mu_2$

Test Statistic

The test statistic follows a standard normal distribution:

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where:

- $\bar{X}_1, \bar{X}_2$ = sample means
- μ_1, μ_2 = population means (under H_0 , $\mu_1 - \mu_2 = 0$)
- σ_1, σ_2 = population standard deviations
- n_1, n_2 = sample sizes

Decision Rule

- Compare the test statistic Z with the critical value from the standard normal table.
- Reject H_0 if:
 - Two-tailed: $|Z| > Z_{\alpha/2}$
 - Right-tailed: $Z > Z_{1-\alpha}$
 - Left-tailed: $Z < -Z_{1-\alpha}$

Example

Problem: A researcher wants to compare the average test scores of two schools. A sample of 40 students from School A has an average score of 78 with a known population standard deviation of 10. A sample of 35 students from School B has an average score of 74 with a known population standard deviation of 8. Test at $\alpha = 0.05$ whether the mean scores are different.

Solution Steps

Step 1: Define Hypothesis

- $H_0 : \mu_1 = \mu_2$
- $H_1 : \mu_1 \neq \mu_2$

Step 2: Compute Test Statistic

$$Z = \frac{(78 - 74) - 0}{\sqrt{\frac{10^2}{40} + \frac{8^2}{35}}}$$

$$Z = \frac{4}{\sqrt{\frac{100}{40} + \frac{64}{35}}}$$

$$Z = \frac{4}{\sqrt{2.5 + 1.8286}} = \frac{4}{\sqrt{4.3286}} = \frac{4}{2.08} = 1.92$$

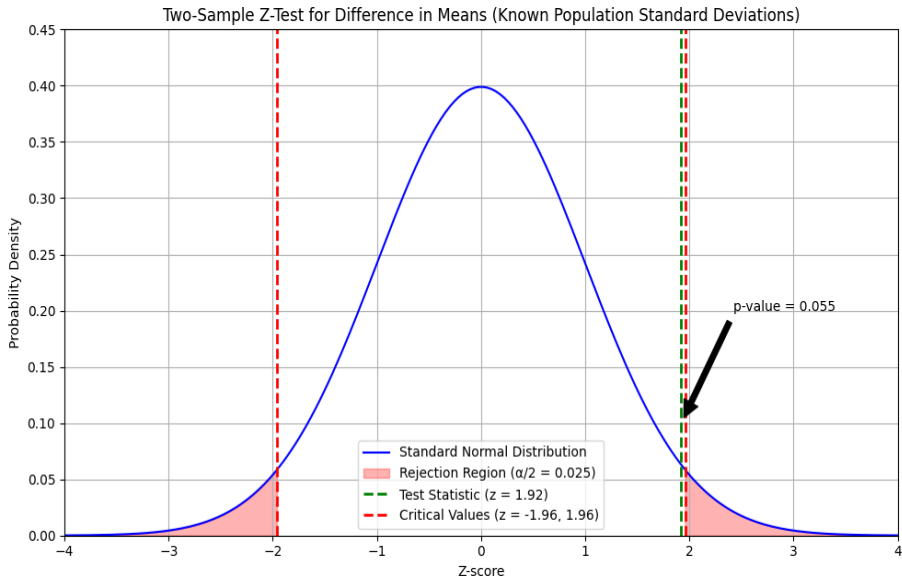
Step 3: Find Critical Value

- For $\alpha = 0.05$, two-tailed test: $Z_{\alpha/2} = 1.96$

Step 4: Decision

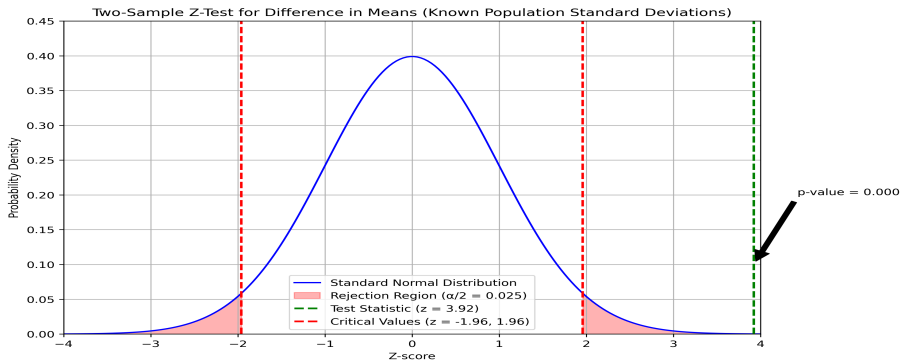
- Since $|1.92| < 1.96$, we fail to reject H_0 .
- Conclusion: There is no significant difference in the mean test scores of the two schools.

Two sample Z test Test



Two sample Z Test

A company wants to compare the average sales performance of two regions. Region 1 has a population mean sales of 200, and Region 2 has 190. The standard deviations for both regions are known to be 15 and 10, respectively. A sample of 50 sales records is taken from each region. Test at a 95% confidence level whether the mean sales are significantly different between the two regions.



Two Sample T-Test

Two-Sample T-Test

Two-Sample T-Test is used to determine whether there is a significant difference between the means of two independent samples when population Standard deviations are unknown.

Applications: Comparing treatment effects, analyzing test scores, evaluating manufacturing processes, etc.

Why Use a Two-Sample T-Test?

- Used when comparing the means of two independent groups.
- Appropriate when population standard deviations are unknown.
- Suitable for small sample sizes ($n_1, n_2 < 30$) if normality is assumed.

What Are Two-Sample Tests?

- Used to compare two population means or proportions.
- Answer the question: **Is there a significant difference between two groups?**
- Two main types:
 - 1 Independent Samples
 - 2 Matched Pairs (Dependent Samples)

Independent Samples: Overview

- Samples come from two unrelated populations.
- Common examples:
 - Male vs Female performance
 - Treatment vs Control group.
 - Test scores in 2 different schools.
- No pairing between individual observations.
- Data in Group A is not related/does not affect data in Group B.
- Can test difference in means as well as difference in proportions.

Independent Samples: Hypotheses and Test

Hypotheses (for means)

$$H_0 : \mu_1 = \mu_2 \quad \text{vs} \quad H_A : \mu_1 \neq \mu_2 \quad (\text{or } <, >)$$

Test Statistic (Equal variances)

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \quad s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}, \quad df = n_1 + n_2 - 2$$

- If variances are unequal, use Welch's t-test.

Welch's t-test: Unequal Variances

- Used when comparing two means from independent samples **with unequal variances**.
- Does not assume equal population variances ($\sigma_1^2 \neq \sigma_2^2$).

Test Statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Degrees of Freedom (Welch–Satterthwaite equation)

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

Matched Pairs: Overview

- Also called **paired samples** or **dependent samples**.
- Each observation in one sample is paired with a specific observation in the other.
- Common examples:
 - Before-and-after measurements
 - Twin studies
- Can only test difference means.

Matched Pairs: Hypotheses and Test

Hypotheses

$$H_0 : \mu_D = 0 \quad \text{vs} \quad H_A : \mu_D \neq 0$$

where μ_D is the mean of the differences.

Test Statistic

$$t = \frac{\bar{d}}{s_D / \sqrt{d}}, \quad \text{df} = n - 1$$

- Simple one-sample t-test on the differences.

When to Use Which?

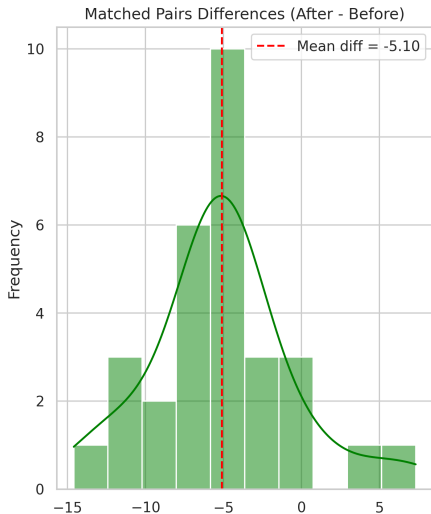
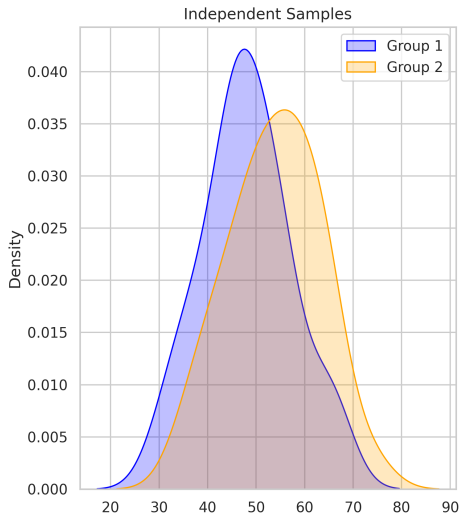
- Use **independent sample test** when:
 - Groups are unrelated.
 - No natural pairing.
- Use **matched pairs test** when:
 - Each subject is measured twice.
 - Observations are naturally linked.

Comparison Table

Feature	Independent Samples	Matched Pairs
Data Structure	Unrelated groups	Paired data
Hypothesis	$\mu_1 = \mu_2$	$\mu_D = 0$
Test	Two-sample t-test	One-sample t-test on D
Examples	Drug A vs Drug B	Before vs After

Two sample Test

- Independent: 2 separate distributions, no link.
- Matched: Pairs of values (like x-y plots), focusing on their difference.



Assumptions

- The two samples are independent.
- The populations are normally distributed (especially important for small samples).
- Population variances are unknown but may be assumed equal or unequal.
- Each sample is randomly selected.

Hypothesis Statements

Two-tailed Test:

- $H_0 : \mu_1 = \mu_2$ (The population means are equal.)
- $H_1 : \mu_1 \neq \mu_2$ (The population means are different.)

One-tailed Tests:

- Right-tailed: $H_1 : \mu_1 > \mu_2$
- Left-tailed: $H_1 : \mu_1 < \mu_2$

Test Statistic

The test statistic follows a t-distribution:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where:

- $\bar{X}_1, \bar{X}_2$ = sample means
- s_1, s_2 = sample standard deviations
- n_1, n_2 = sample sizes
- $\mu_1 - \mu_2 = 0$ under H_0

Degrees of freedom are estimated using:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}$$

Decision Rule

- Compare the test statistic t with the critical value from the t-table.
- Reject H_0 if:
 - Two-tailed: $|t| > t_{\alpha/2, df}$
 - Right-tailed: $t > t_{1-\alpha, df}$
 - Left-tailed: $t < -t_{1-\alpha, df}$

Example

Problem: A company tests a new training program. A sample of 15 employees trained with the new program has an average productivity score of 85 with a standard deviation of 5. Another sample of 12 employees trained with the old program has an average score of 80 with a standard deviation of 6. Test at $\alpha = 0.05$ whether the new program is more effective.

Solution Steps

Step 1: Define Hypothesis

- $H_0 : \mu_1 = \mu_2$
- $H_1 : \mu_1 > \mu_2$

Step 2: Compute Test Statistic

$$t = \frac{(85 - 80) - 0}{\sqrt{\frac{5^2}{15} + \frac{6^2}{12}}}$$

$$t = \frac{5}{\sqrt{\frac{25}{15} + \frac{36}{12}}}$$

$$t = \frac{5}{\sqrt{1.67 + 3}} = \frac{5}{2.29} = 2.18$$

Step 3: Find Critical Value

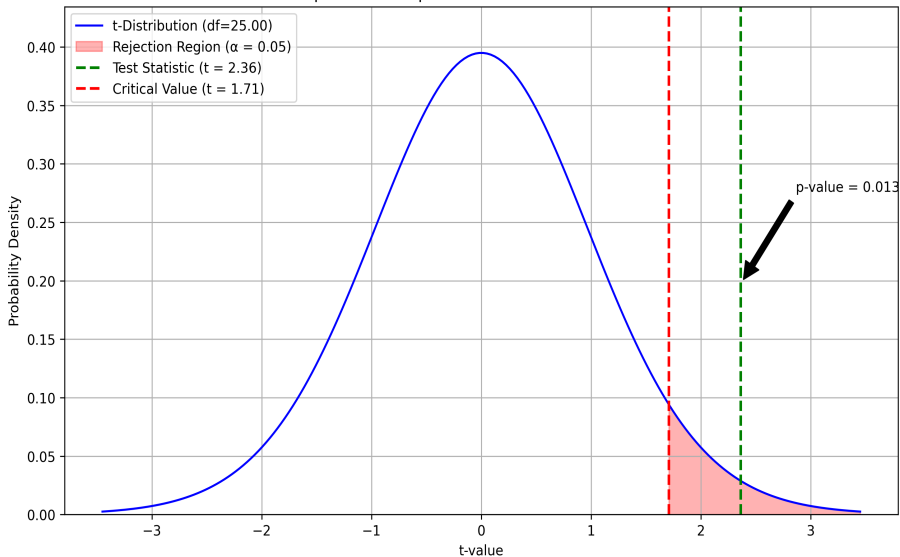
- Degrees of freedom are estimated using the formula.
- Using a t-table, for $df \approx 20$ and $\alpha = 0.05$ (one-tailed test),
 $t_{0.05,20} = 1.725$.

Step 4: Decision

- Since $t = 2.18 > 1.725$, we reject H_0 .
- Conclusion: There is significant evidence that the new training program improves productivity.

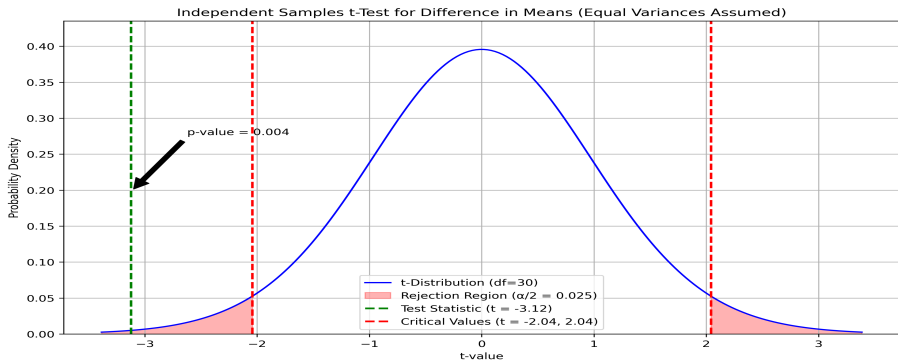
Two sample t Test

Independent Samples t-Test for Difference in Means



Two sample t Test

A study is conducted to compare the average test scores of two different teaching methods. A random sample of 16 students from Method A has a mean score of 75 with a variance of 16, and a random sample of 16 students from Method B has a mean score of 80 with a variance of 25. Test at a 95% confidence level whether there is a significant difference in the mean scores of the two methods. (assume equal variance)



Test Statistic

The paired t-test uses the mean of the differences ($\bar{d}$):

$$t = \frac{\bar{d} - \mu_d}{s_d / \sqrt{n}}$$

where:

- $\bar{d}$ = mean of the differences between paired values
- s_d = standard deviation of the differences
- n = number of pairs
- $\mu_d = 0$ under H_0

Example

Problem: A researcher tests a diet program on 10 individuals. Their weights (in kg) before and after are recorded. Test at $\alpha = 0.05$ whether the program is effective.

Data:

Participant	Before	After	Difference (d_i)
1	80	76	4
2	85	82	3
3	78	75	3
4	90	88	2
5	76	72	4
6	82	79	3
7	88	83	5
8	92	89	3
9	75	73	2
10	84	80	4

Solution

Step 1: Compute Differences

$$d_i = \text{Before} - \text{After}$$

Step 2: Compute Mean and Standard Deviation of Differences

$$\bar{d} = 3.2, \quad s_d = 1.75$$

Step 3: Compute Test Statistic

$$t = \frac{3.2 - 0}{1.75/\sqrt{10}} = \frac{3.2}{0.553} = 5.78$$

Step 4: Decision

- For $df = 9$, $t_{0.05,9} = 2.262$.
- Since $5.78 > 2.262$, we reject H_0 .
- Conclusion: The diet program significantly reduced weight.

Two-Proportion Z-Test

Two-Proportion Z-Test

Two-Proportion Z-Test is used to compare two population proportions to determine whether there is a significant difference between them.

Applications: Medical studies, A/B testing in marketing, election polling, etc.

Why Use a Two-Proportion Z-Test?

- Used when comparing the proportion of successes in two independent samples.
- Suitable for large sample sizes where normal approximation holds.
- Helps determine if a difference in proportions is statistically significant.

Assumptions

- The samples are independent.
- Data is categorical.
- Each sample follows a binomial distribution.
- The normal approximation is valid:
 - $n_1 p_1 \geq 5, n_1(1 - p_1) \geq 5$.
 - $n_2 p_2 \geq 5, n_2(1 - p_2) \geq 5$.

Test Statistic

The test statistic follows a standard normal distribution:

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{p(1-p) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

where:

- $\hat{p}_1 = \frac{x_1}{n_1}$ and $\hat{p}_2 = \frac{x_2}{n_2}$ are sample proportions.
- p is the pooled proportion: $p = \frac{x_1 + x_2}{n_1 + n_2}$.
- n_1, n_2 are sample sizes, and x_1, x_2 are the number of successes.

Example

Problem: A company tests two different advertisements. Out of 200 people who saw Ad A, 80 made a purchase. Out of 250 people who saw Ad B, 100 made a purchase. Test at $\alpha = 0.05$ whether Ad B performs significantly better.

Step 1: Define Hypotheses

- $H_0 : p_1 = p_2$ (No difference in effectiveness)
- $H_1 : p_1 < p_2$ (Ad B is more effective)

Step 2: Compute Sample Proportions and Pooled Proportion

$$\hat{p}_1 = \frac{80}{200} = 0.40, \quad \hat{p}_2 = \frac{100}{250} = 0.40$$

$$p = \frac{80 + 100}{200 + 250} = 0.40$$

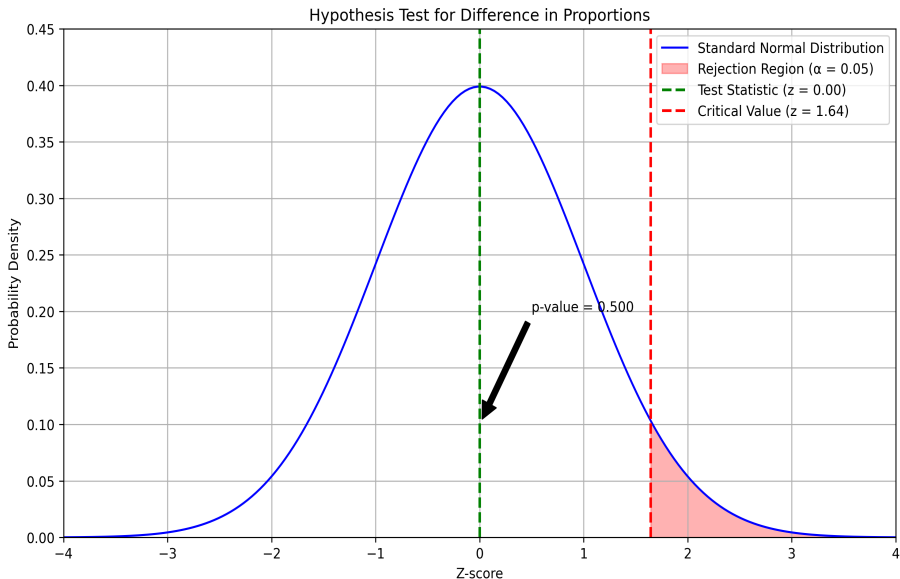
Step 3: Compute Test Statistic

$$Z = \frac{(0.40 - 0.40)}{\sqrt{0.40(1 - 0.40) \left(\frac{1}{200} + \frac{1}{250} \right)}} = 0$$

Step 4: Decision

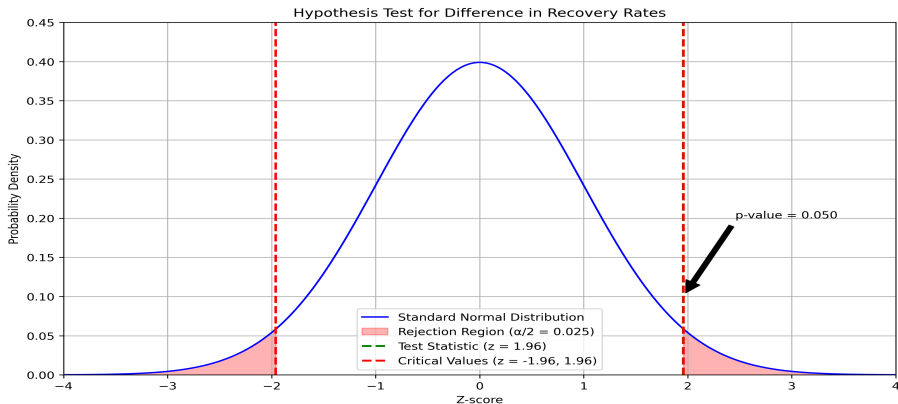
- Critical value for one-tailed test at $\alpha = 0.05$ is 1.645.
- Since $Z = 0 < 1.645$, we fail to reject H_0 .
- Conclusion: No significant difference in ad effectiveness.

Two proportion Test



Two proportion Test

A researcher wants to compare the effectiveness of two different drugs. In a clinical trial, Drug A was given to 200 patients, and 120 of them recovered. Drug B was given to 180 patients, and 90 of them recovered. Test at a 95% confidence level whether there is a significant difference in the recovery rates between the two drugs.



Two Variance F-Test

Two-Variance F-Test

Two-Variance F-Test is used to compare the variances of two independent populations to determine if they are significantly different.

Applications: Quality control, financial risk analysis, comparing variability in different groups.

Why Use a Two-Variance Test?

- Used when assessing variability between two independent samples.
- Helps in determining the assumption of equal variances for t-tests.
- Important in fields like manufacturing and medical research.

Assumptions

- The two samples are independent.
- Each sample follows a normal distribution.
- The ratio of variances follows an F-distribution.

Test Statistic

The test statistic follows an F-distribution:

$$F = \frac{s_1^2}{s_2^2}$$

where:

- s_1^2 and s_2^2 are the sample variances.
- The larger variance is placed in the numerator to ensure $F \geq 1$.
- Degrees of freedom: $df_1 = n_1 - 1$, $df_2 = n_2 - 1$.

Example

Problem: A manufacturer tests the variability in product weight from two production lines. A sample of 10 products from line A has a variance of 4.5, while a sample of 12 products from line B has a variance of 2.8. Test at $\alpha = 0.05$ whether the variances are significantly different.

F-Test for Equality of Variances

- **Null hypothesis:** $H_0 : \sigma_1^2 = \sigma_2^2$
- **Alternative hypothesis:** $H_1 : \sigma_1^2 \neq \sigma_2^2$
- **F-statistic calculation:**

$$F = \frac{s_1^2}{s_2^2} = \frac{4.5}{2.8} \approx 1.6071$$

- **Degrees of freedom:**

$$df_1 = n_1 - 1 = 10 - 1 = 9 \quad \text{and} \quad df_2 = n_2 - 1 = 12 - 1 = 11$$

- **Critical F-values:**

$$F_{\alpha/2} = 0.2915 \quad \text{and} \quad F_{1-\alpha/2} = 3.4384$$

- **p-value:**

$$p\text{-value} \approx 1.6452$$

- **Conclusion:** Fail to reject the null hypothesis. The variances are not significantly different.

Step 1: Define Hypotheses

- $H_0 : \sigma_1^2 = \sigma_2^2$ (No difference in variances)
- $H_1 : \sigma_1^2 \neq \sigma_2^2$ (Variances are different)

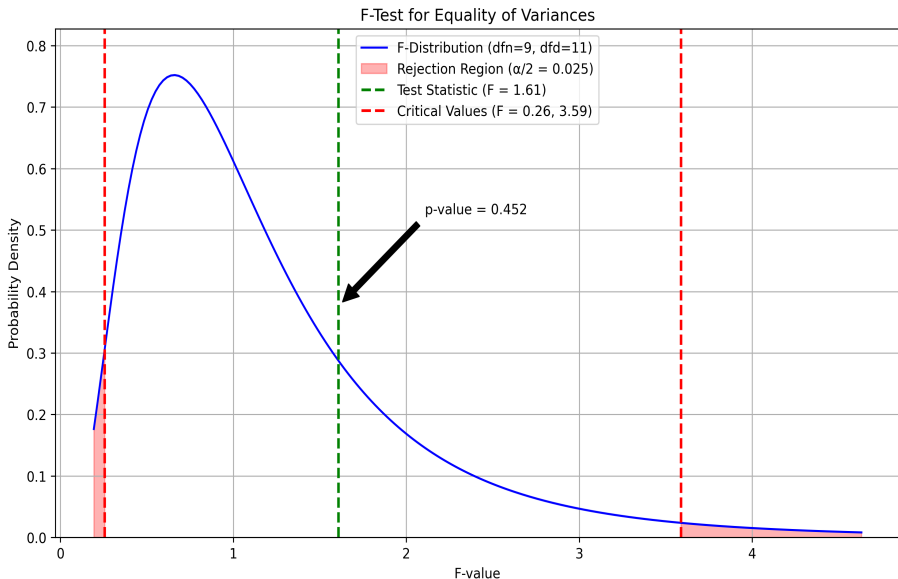
Step 2: Compute Test Statistic

$$F = \frac{4.5}{2.8} = 1.607$$

Step 3: Decision

- Critical value for $F_{(9,11)}$ at $\alpha = 0.05$ (two-tailed) is approximately 3.29.
- Since $1.607 < 3.29$, we fail to reject H_0 .
- Conclusion: No significant difference in variances.

Two variance Test



ANOVA

What is ANOVA?

- ANOVA stands for **Analysis of Variance**.
- It is a statistical method used to **test the difference between two or more group means**.
- Developed by Ronald Fisher in the early 20th century.
- Used primarily to analyze experimental data.
- **Applications:** Medical studies, agriculture, business decision-making, industrial quality control.

Why Use ANOVA?

- Compares means across multiple groups simultaneously.
- Controls Type I error rate better than multiple t-tests.
- Tests for overall significance: is there at least one group mean different?
- Helps identify sources of variability in the data.
- For example, comparing 4 groups would require 6 pairwise t-tests.
- Each test at $\alpha = 0.05$ leads to a compounded error rate.
- ANOVA provides a single test to maintain the overall error rate.

Types of ANOVA

- **One-Way ANOVA:**

- Compares means of three or more independent groups based on one categorical independent variable.
- Example: Comparing test scores across three different teaching methods.

- **Two-Way ANOVA:**

- Examines the effect of two independent variables simultaneously and their potential interaction effect.
- Example: Studying the effect of diet type and exercise level on weight loss.

- **Repeated Measures ANOVA:**

- Used when the same subjects are measured multiple times under different conditions.
- Example: Measuring the blood pressure of patients before, during, and after a treatment.

- **Factorial ANOVA:**

- An extension of Two-Way ANOVA that allows for multiple independent variables and their interactions with one dependent variable.
- Example: Studying the impact of age, gender, and medication on blood pressure.

Why Use ANOVA?

- Extends the t-test to multiple groups.
- Prevents increased Type I error due to multiple comparisons.
- Determines whether observed differences in sample means are due to actual differences or random variation.

Assumptions

- The samples are independent.
- Each group follows a normal distribution.
- Homogeneity of variance: all groups have equal variances.

Steps in One-Way ANOVA

- 1 State the hypotheses:
 - H_0 : All group means are equal.
 - H_a : At least one group mean is different.
- 2 Calculate the group means and overall mean.
- 3 Compute the between-group and within-group variability.
- 4 Calculate the F-statistic:

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$$

- 5 Compare F with critical value or use p-value.
- 6 Draw conclusion.

One-Way ANOVA Example

- Suppose we test the effect of 3 diets on weight loss: Diet A, Diet B, and Diet C.
- Subjects are randomly assigned to one of the diets.
- After 4 weeks, their weight loss is recorded.
- Use ANOVA to test if the average weight loss differs among the 3 diets.

Assumptions of ANOVA

- Independence of observations.
- Normally distributed populations.
- Homogeneity of variances (equal variances).

One-Way ANOVA Example

- **Research Question:** Do teaching methods affect student test scores?
- **Groups(Treatments):**
 - Method A: 85, 90, 88
 - Method B: 78, 75, 80
 - Method C: 82, 85, 87

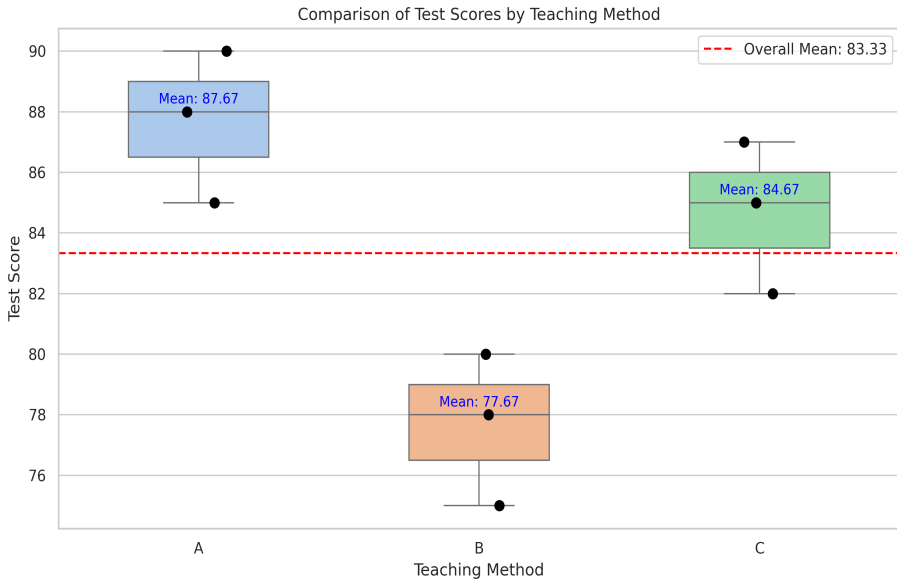
Step 1: Hypotheses

- **Null Hypothesis (H_0):** All group means are equal.
- **Alternative Hypothesis (H_1):** At least one group mean is different.

Step 2: Group Means and Overall Mean

- A: $\frac{85+90+88}{3} = 87.67$
- B: $\frac{78+75+80}{3} = 77.67$
- C: $\frac{82+85+87}{3} = 84.67$
- Overall Mean: $\bar{X} = \frac{85+90+88+78+75+80+82+85+87}{9} = 83.44$

Group Means and Overall Mean



Step 3: Sums of Squares

Between- Group (Treatment) Sum of Squares (SSB):

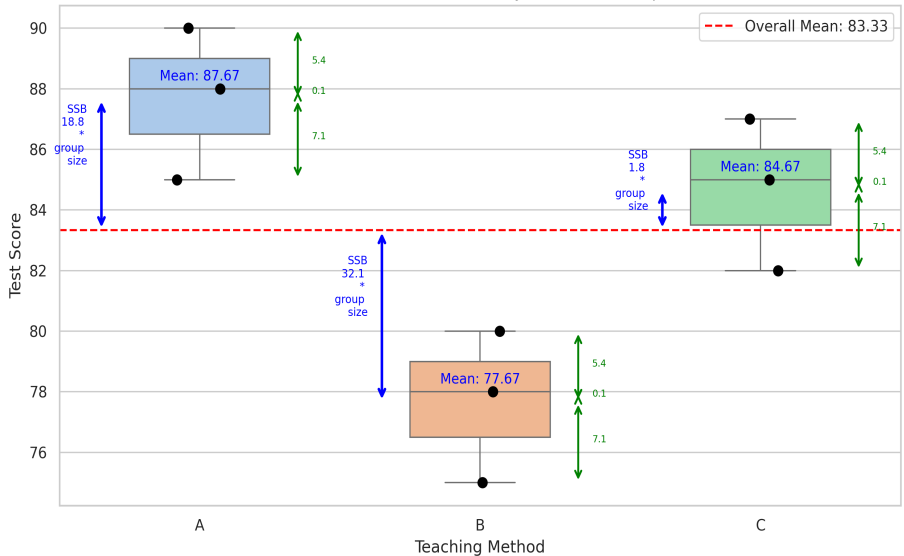
$$SSB = 3((87.67 - 83.44)^2 + (77.67 - 83.44)^2 + (84.67 - 83.44)^2) = 158.25$$

Within- Group (Treatment) Sum of Squares (SSW):

- Group A: $(85 - 87.67)^2 + (90 - 87.67)^2 + (88 - 87.67)^2 = 12.67$
- Group B: $(78 - 77.67)^2 + (75 - 77.67)^2 + (80 - 77.67)^2 = 12.67$
- Group C: $(82 - 84.67)^2 + (85 - 84.67)^2 + (87 - 84.67)^2 = 12.67$
- Total SSW = 38.

Between and Withing Group Variance

SSB and SSW Visualized Clearly Outside the Boxplot



Step 4: Mean Squares and F-Ratio

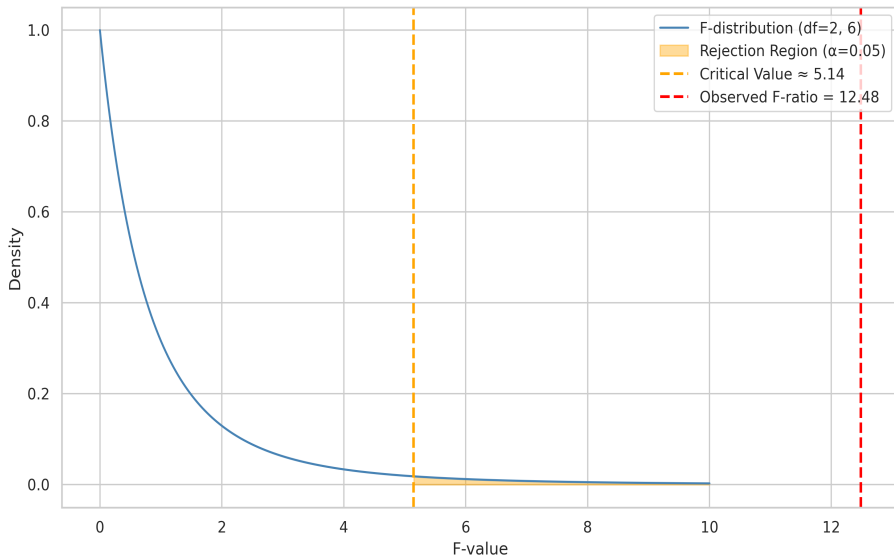
- **df Between** $(k-1) = 3 - 1 = 2$
- **df Within** $(N-k) = 9 - 3 = 6$
- **MST** $= \frac{158.25}{2} = 79.13$
- **MSE** $= \frac{38.01}{6} = 6.34$
- **F-Ratio** $= \frac{79.13}{6.34} = 12.48$

Step 5: Decision and Conclusion

- Critical F-value at $\alpha = 0.05$, $df_1 = 2$, $df_2 = 6$ is approximately 5.14
- Since $12.48 > 5.14 \Rightarrow \text{Reject } H_0$
- **Conclusion:** There is a significant difference in test scores among teaching methods.

ANOVA

F-Distribution with Observed F-Ratio and Critical Value



ANOVA Components

- **Total Sum of Squares (SST):** Measures overall variation in the data.

$$SST = \sum (X_{ij} - \bar{X}_{\text{overall}})^2$$

- **Between-Group Sum of Squares (SSB):** Measures variation due to differences among group means.

$$SSB = \sum n_i (\bar{X}_i - \bar{X}_{\text{overall}})^2$$

- **Within-Group Sum of Squares (SSW):** Measures variation within each group.

$$SSW = \sum (X_{ij} - \bar{X}_i)^2$$

Mean Squares and F-Statistic

- **Mean Square Between (MSB):**

$$MSB = \frac{SSB}{df_B} = \frac{SSB}{k - 1}$$

- **Mean Square Within (MSW):**

$$MSW = \frac{SSW}{df_W} = \frac{SSW}{N - k}$$

- **F-Statistic:**

$$F = \frac{MSB}{MSW}$$

Example 1

Problem: A researcher compares the effectiveness of three diets. The weight loss (in kg) of participants on each diet is recorded. Test at $\alpha = 0.05$ whether there is a significant difference.

	Diet Group	Weight Loss (kg)
Data:	A	5, 7, 6, 8, 7
	B	6, 9, 7, 10, 8
	C	4, 6, 5, 7, 6

Example 2

Problem: A study examines the effectiveness of three different fertilizers on plant growth (in cm). The growth measurements are recorded below. Test at $\alpha = 0.05$ if there is a significant difference.

Data:

Fertilizer	Plant Growth (cm)
X	12, 15, 14, 16, 15
Y	18, 20, 19, 22, 21
Z	10, 12, 11, 13, 12

Solution for Example 2

Step 1: Compute Group Means and Overall Mean

$$\bar{X}_X = 14.4, \quad \bar{X}_Y = 20, \quad \bar{X}_Z = 11.6, \quad \bar{X}_{\text{overall}} = 15.33$$

Step 2: Compute SST, SSB, and SSW

$$SST = 64.8, \quad SSB = 54.93, \quad SSW = 9.87$$

Step 3: Compute MSB and MSW

$$MSB = \frac{54.93}{2} = 27.47$$

$$MSW = \frac{9.87}{12} = 0.82$$

Step 4: Compute F-Statistic

$$F = \frac{MSB}{MSW} = \frac{27.47}{0.82} = 33.5$$

Step 5: Decision

- Critical value for $F_{(2,12)}$ at $\alpha = 0.05$ is 3.89.
- Since $33.5 > 3.89$, we reject H_0 .
- Conclusion: There is a significant difference between the fertilizers.

Goodness Of Fit

Goodness of fit measures how well a statistical model fits a given dataset.

- Evaluates the difference between observed and expected values.
- Helps determine if a model is appropriate for the data.

Why Goodness of Fit?

- Ensures the model represents the data accurately.
- Helps in hypothesis testing.
- Avoids incorrect conclusions from poorly fitting models.

Methods of Goodness of Fit

Common methods include:

- **Chi-Square Test**
- **Kolmogorov-Smirnov Test**
- **Anderson-Darling Test**
- **R-squared for Regression Models**

Chi-Square Test

Used for categorical data:

- Compares observed and expected frequencies.
- Formula:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Example: Chi-Square Test

Suppose we survey customers about their product preferences:

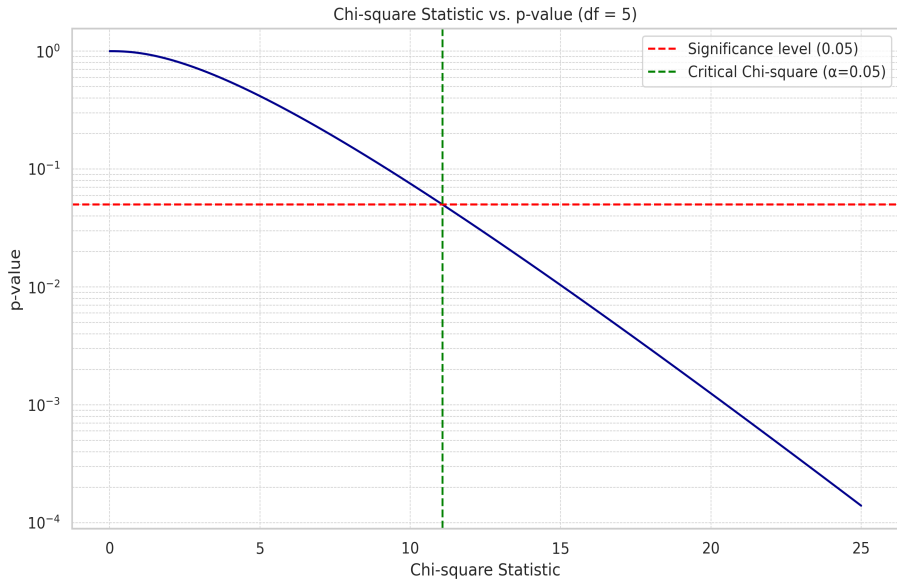
- Observed counts: (50, 30, 20)
- Expected counts: (40, 40, 20)

Applying Chi-Square Formula:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Decision Rule: If χ^2 is large, reject the model fit.

Chi Square Test Statistic vs p-value



Conclusion

- Goodness of fit helps validate models before making decisions.
- Different tests apply based on data type.
- A good fit increases confidence in statistical conclusions.

Summary of One Sample Tests

One-Sample Tests

- **One Sample z Test:** Known population standard deviation; test mean vs a value
- **One Sample t Test:** Unknown population standard deviation; small sample size
- **One Proportion Test:** Tests if sample proportion differs from hypothesized value
- **One Variance Test:** Tests if sample variance differs from a known value (Chi-Square)

Summary of Two Sample Tests

Two-Sample Tests

- **Two Sample z Test:** Compare two means, large n , known population variances
- **Two Sample t Test:** Compare means of two independent samples (equal/unequal variance)
- **Paired t Test:** Same subjects tested twice (before/after), tests mean of differences
- **Two Proportion Test:** Compares proportions from two independent groups
- **Two Variance Test:** Compares variability (F-test) between two groups

When and Where to Use These Tests

- **Use z-test (mean):** If population standard deviation is known and $n \geq 30$
- **Use t-test (mean):** If population standard deviation is unknown or small sample
- **Use paired t-test:** If same subject is measured twice (before vs after scenarios)
- **Use proportion test:** If the outcome is categorical (e.g., pass/fail)
- **Use variance test:** To compare variability or consistency between groups
- **Use ANOVA:** To compare means across 3 or more independent groups

THANK YOU